

GENERAL SPECIFICATION G4.4

Pumps

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G4.4 PUMPS

G4.4.1 Definitions

- (a) This specification covers the general requirement for both water and used water (wastewater) pumping requirement. The specific requirements for each particular application are described in the Particular Specification.
- (b) In the event of any conflict between this General Specification and the Particular Specification, the latter shall take precedence and the Contractor shall write to the S.O. for clarification.
- (c) For the purpose of the Works, the following definitions shall apply:
 - (i) Design duty - The total head to be developed and the quantity of fluid to be discharged per unit time when the pump is running at rated speed;
 - (ii) Static head - The difference between free water surface level on the suction side of the pump and the water surface level on delivery side or highest point on delivery side, whichever is higher;
 - (iii) External friction head - The head required to overcome friction external to the Works and the velocity head at the outlet of the pumping main;
 - (iv) External plant losses - The friction losses in all pipework and valves included in the Works;
 - (v) Internal plant losses - The friction losses in the pump suspension main and head bend up to the delivery flange of vertical wet well pumps;
 - (vi) Total dynamic head (TDH) - The sum of (ii), (iii), (iv) and (v);
 - (vii) Net effective lift - The sum of (ii) and (iii);
 - (viii) NPSH (Net positive suction head) - Net positive suction head available (NPSHA) is the head available at the suction side of pump after subtracting the vapour pressure of the liquid being pumped. Net positive suction head required (NPSHR) is the head at the suction side of pump that causes a 3 % reduction in the TDH; and
 - (ix) Unless otherwise specified, NPSHA margin above NPSHR shall comply with ANSI/HI 9.6.1 American National Standard for Rotodynamic Pumps. The margins for various pump applications are summarised as follows:

Application Description		NPSHA Minimum Requirement
Water pumping	Pumping power less than 75 kW (per stage)	NPSHA value calculated using the ratio of $NPSHA/NPSHR = 1.1$ or $NPSHA - NPSHR = 1.5$ m whichever is greater.
	Pumping power of 75 kW and greater (per stage)	NPSHA value calculated using the ratio of $NPSHA/NPSHR = 1.2$ or $NPSHA - NPSHR = 1.5$ m whichever is greater.

Application Description		NPSHA Minimum Requirement
Wastewater pumping	Pumping power less than 45 kW (per stage)	NPSHA value calculated using the ratio of $NPSHA/NPSHR = 1.2$ or $NPSHA - NPSHR = 1.5$ m whichever is greater.
	Pumping power of 45 kW and greater (per stage)	NPSHA value calculated using the ratio of $NPSHA/NPSHR = 1.3$ or $NPSHA - NPSHR = 1.5$ m whichever is greater.
Process chemical pumping		NPSHA value calculated using the ratio of $NPSHA/NPSHR = 1.2$ or $NPSHA - NPSHR = 1.0$ m whichever is greater.
Slurry pumping		NPSHA value calculated using the ratio of $NPSHA/NPSHR = 1.1$ or $NPSHA - NPSHR = 0.6$ m whichever is greater.
General industrial pumping		NPSHA value calculated using the ratio of $NPSHA/NPSHR = 1.1$ or $NPSHA - NPSHR = 1.0$ m whichever is greater.

G4.4.2 Submittals

- (a) The Contractor shall submit complete pump and pump motor technical schedules/datasheets, fabrication and assembly drawings, together with detailed specifications and data covering materials, drive unit, parts, devices, and accessories forming a part of the equipment furnished. The data and specifications for each unit shall include but shall not be limited to the following:
- (i) For the Pump:
- 1) Manufacturer/Brand;
 - 2) Type and model;
 - 3) Rotational speed;
 - 4) Net weight of pump unit;
 - 5) Net weight with baseplate and coupling;
 - 6) Size of suction and discharge flange;
 - 7) Complete performance curves for all specified operating points showing capacity versus head, NPSH required, pump efficiency, and pump input power;
 - 8) Type of coupling;
 - 9) Cooling water/flushing water requirements (flowrate, pressure) for bearings and mechanical seals;
 - 10) Base and anchor bolt details;
 - 11) Details parts and components list with materials of construction;
 - 12) Data on shop painting;

- 13) General arrangement drawings;
 - 14) Detailed fabrication drawings for baseplate;
 - 15) Recommended spare parts list;
 - 16) Operation and Maintenance manual;
 - 17) Erection/Installation Instructions; and
 - 18) Type of seals.
- (ii) For the Motors:
- 1) Name of manufacturer;
 - 2) Type and model;
 - 3) Type of bearings, method of lubrication, lubrication media flowrate and pressure;
 - 4) Motor cooling method and cooling media flowrate and pressure;
 - 5) Rated size of motor, kW, and service factor;
 - 6) Insulation class and temperature rise;
 - 7) Full load rotational speed and number of poles;
 - 8) Net weight;
 - 9) Efficiency at full load and rated pump condition;
 - 10) Full load current;
 - 11) Locked rotor current; and
 - 12) Any other documents as required in Electrical Specifications.
- (iii) For Control Panel and Components:
- 1) Name of manufacturer;
 - 2) Type and model;
 - 3) Enclosure rating;
 - 4) Dimensions of complete panel;
 - 5) Electrical schematics and wiring diagram;
 - 6) Liquid level sensors with mounting details and cable lengths, and pump controls;
 - 7) Published descriptive data on each item of equipment and all accessories indicating all specific characteristics; and
 - 8) Any other documents as required in Instruments & Control Specifications.
- (iv) For the adjustable frequency drives: – as specified in the Electrical Specifications.

G4.4.3 General

- (a) Each pumping unit shall be complete with a pump, electric motor, coupling, coupling guard, pedestal or motor stool (where applicable), anchor bolts, and other appurtenances specified or otherwise required for proper operation, all mounted on a common baseplate.
- (b) The Contractor shall obtain and submit the pump manufacturer's guarantee that the combined performance of pump and motor set as a package including the sizing and sourcing of the suitable motor to meet the requirements.
- (c) The proposed pump shall be selected taking into consideration its suitability and fit for use in the environment and service conditions intended for.
- (d) All motor enclosures shall be compatible with the area of installation and the intended use.
- (e) Unless otherwise specified, the pump and motor shall be mounted on a common baseplate assembly or when required a separate motor stool for mounting of the drive motor. Baseplate and motor stool shall be mounted on a suitable concrete foundation. The baseplate and/or motor stool shall be able capable of carrying the weight of the pump and/or motor.
- (f) Particular requirements of the pumps required for the contract such as specific pump duty points, pump types, material of construction, etc. are given in the Particular Specification.
- (g) The safety requirements of the pump unit shall comply with BS EN 809.
- (h) Unless otherwise specified, power supply to the equipment shall be 400 volts, 50 Hz, 3 phase. See Electrical specifications for further requirements and particularly for large pump units requiring higher power supply voltage.
- (i) Pump Design Requirements:
 - (i) All pumps shall have a non-overloading performance characteristics. The efficiency shall be high at the design duty point and remain at a reasonable high level over the duty range of the pumping system.
 - (ii) The pump head/flowrate characteristic shall be stable at all rates of flow between closed valve and open valve and shall be steep enough to permit satisfactory operation in parallel under all conditions specified.
 - (iii) The head versus flow rate characteristics curve shall be continuously falling from a maximum at closed valve head.
 - (iv) The power delivered to the pump coupling shall be adequate for the full characteristic range from the closed valve conditions to the run-out point.
 - (v) The pump efficiency shall be maintained over the whole of the specified duty range, even if this necessitates the use of a larger motor to provide the peak power demand. For vertical spindle suspended pumps the efficiency shall take into account shaft and rising main losses up to and including the pump discharge bend, and the pump coupling.

- (vi) Water velocities in the suction piping connected to a pump shall not exceed 2.4 m/s. Unless otherwise specified, velocities in the discharge branch shall not exceed the manufacturer's recommendation for flow in valves and fittings or be within the range of 0.9 to 2.4 m/s, whichever is the lower when the pump is operating within its specified duty range and application. Within this range there shall be no discernible noise due to hydraulic turbulence or cavitation within either the pump or its associated pipework and valves. In the event where discharge velocities exceed 4.5 m/s in the duty range, the use of profiled taper pipes will be permitted. Taper pipes shall be specifically designed to progressively reduce velocity with profiled internal contours, which contain no steps or acute angle from which turbulence or cavitation can occur. Taper pipes shall be considered as part of the pump and be included in all performance calculations and tests.
- (vii) Pumps for raw water applications shall be capable of passing 75 mm diameter solids unless otherwise specified in the Particular Specification and shall be designed to avoid possible choking by weeds or other tough sinuous materials.
- (viii) Pumps for sewage, used water (wastewater) and sludge application shall be capable of passing minimum of 120 mm diameter solids unless otherwise specified in the Particular Specification and shall be designed to avoid possible choking by screenings, rags, plastic debris, stringy materials, hairs and fibres.
- (ix) For sludge applications, selection of the type of pump and impeller shall take into consideration type and content of sludge being pumped.
- (x) In general, the speed of main pumps shall not exceed 1,500 RPM unless specified.
- (xi) For pumps condition monitoring and protection system, refer to the relevant sections of this specification, the Particular Specification and the electrical General Specification.
- (xii) Motor ratings shall take account of the maximum power requirement of the pumps and their driving arrangements over the entire design speed range, and shall include a margin of not less than 15 % in excess of the maximum power required in any possible operating condition.
- (j) Pump Construction:
 - (i) Unless otherwise specified, the material selection shall be based on the specified operating environment, application, pumped medium, pump duty and the expected service lives of the specified pump unit components.
 - (ii) Where there is contact or exposure to liquids in which chlorides are present, the wetted parts and components of the pumps shall be manufactured of materials guaranteed against corrosion and wear with chlorides concentration as follows:

Liquid in contact	Chloride concentration limit
Saturated brine	At least 160, 000 mg/l
Seawater	At least 19, 000 mg/l
All other liquid with chlorides present	At least 10, 000 mg/l

- (iii) Unless otherwise specified, parts and materials of equipment in contact with seawater shall be fabricated in Super Duplex stainless steel grade SAF2507 or equivalent.
- (iv) All Super Duplex shall have a minimum Pitting Resistance Equivalent Number (PREN) of 40 and Crevice Factor (CF) above 38 as calculated as follow:
$$\text{PREN} = \text{Cr (\%)} + 3.3 \times \text{Mo (\%)} + 16 \times \text{N (\%)}$$
$$\text{CF} = \text{Cr (\%)} + 3 \times \text{Mo (\%)} + 15 \times \text{N (\%)}$$
- (v) The casing material shall be designed to withstand any pressure or dynamic loading that can be generated during normal pump operation, including intermittent physical shock loadings caused by solids in flow and shall resist corrosion and abrasion imposed by constituents in the pumped medium. Access ports shall be provided, if specified, to allow the clearance of blockage and the inspection of the rotor drive components.
- (vi) Pump casing shall be hydrostatically pressure tested to a pressure not less than 1.5 times shut-off head of pump plus maximum suction head unless otherwise specified. Duration of pressure test shall be 30 min. unless otherwise specified in the Particular Specification. The impeller shall be dynamically balanced and free of blow holes, projections, cracks and inclusions.
- (vii) Pumps for sewage and sludge application shall have impellers with non-clog characteristics designed for municipal sewage pumping. The impeller shall be suitable for the pumping application as specified. The impeller shall be capable of handling solids, fibrous or stringy materials, heavy sludge and other matter found in the raw sewage. The Contractor shall obtain and furnish the calculations of the pumps' mass moment of inertia from the pump manufacturers upon request by the S.O.. Impellers shall be keyed to the shaft, retained with an expansion ring.
- (viii) Unless otherwise specified, for pumping chemical fluids applications, the pump materials and design, the pipework system including valves and fittings shall be fully compatible with the chemical to be pumped with special consideration to the chemical properties and be fully compliant with the requirements on chemical dosing pumps in this specification.
- (ix) The rotating assemblies shall be statically and dynamically balanced in accordance with BS ISO 21940-11 and designed so that the first critical speed of the pump and its drive is at least 50% higher than the maximum operating speed. Where the rotating assemblies are small and any out of balance forces are negligible, the S.O. may waive this requirement. The Contractor shall obtain and submit the pump manufacturer's statement of confirmation that balancing has been completed. to state whether or not balancing has been completed.
- (x) Pump wear rings shall be easily replaceable.
- (xi) In general, Impellers shall be fully shrouded and secured to the shaft by a key. The impeller retaining nut shall be fitted with a locking device. Shafts shall be stainless steel forgings compatible with the impellers.

- (xii) Pump design shall incorporate means to prevent impeller loosening, dropping off or bearings failure particularly in event of reverse flow by means of impeller locking device or reverse rotation prevention back stop device of impeller or motor or both subject to approval by the S.O..
- (xiii) Unless otherwise specified, all pumps shall be fitted with suction and delivery pressure gauges, automatic combination air valves, casing drains and lifting lugs. Split casing and end suction pumps shall also be fitted with suction compound vacuum-pressure gauges and drain cocks.
- (xiv) Integral flange shall be provided for connection between pump set and pipework. Flanges shall be rated at generally not less than PN 10 unless otherwise specified and shall have drillings to BS EN 1092. A ½" BSP female parallel threaded tapping point shall be provided at the flange peripheral for the reception of pressure tubing. All flanges shall be machined on face and edge and shall be spigotted to facilitate perfect alignment.
- (xv) All casing drains shall be fitted at the lowest point of casing complete with a tundish or piped to the discharge points approved by S.O..
- (xvi) Facilities shall be provided for the removal of trapped air on priming and for draining the pump.
- (xvii) All exposed rotating parts shall be provided with suitable guards.
- (xviii) All chemical dosing pumps shall have provision for flushing water lines with discharges leading to approved drain points. Specific requirements for the dosing pump shall be as described in relevant section of this specification.
- (xix) For the sludge pumps applications, pump shall also include a provision and for the installation of manual or automatic mechanisms for drawing and clearing the excess air trapped inside pump casing prior to pump operation.
- (xx) Unless otherwise specified in this document, or in the particular specification, the minimum materials of construction shall be as follows:

Casing and Casing Covers	Iron; Cast iron, ASTM A48 Class 30
Impeller (for centrifugal pumps)	Iron, Cast Iron, ASTM A48 Class 30
Shaft (for centrifugal pumps)	Carbon steel AISI 1045

(k) Mechanical Seal:

- (i) Pumps shall be fitted with cartridge type self-aligning mechanical seals with a proven record of satisfactory running over a long period when fitted to the pump in question. Mechanical seal shall be of the self-adjusting non-water cooled type unless specified otherwise. They shall be designed for easy adjustment and seal removal with the wear surfaces removable to facilitate inspection or ease of repair without the need for any special tool.
- (ii) Split type mechanical seal shall be furnished where specified in the Particular Specification especially for pumps installed in confined space situations or deep chambers.

- (iii) For sewage and sludge application, mechanical seal shall be suitably designed for handling abrasive medium.
 - (iv) All metal parts of seal shall be Type SS 316. Material for the rotating face shall be plated tungsten or titanium or approved equal. Material for stationary face shall be carbon graphite or fluorocarbon or approved equal.
 - (v) The 'O' ring material shall be fluorocarbon or fluoric rubber or approved equivalent suitable for use in the pumping media, it shall be designed to sustain flexibility, absorb shaft movements and take up wear ensuring perfect sealing contact. The helical springs shall be of equivalent or better corrosion resistant materials and shall provide consistent face loading at all operating conditions.
 - (vi) Effective means shall be provided for collecting seal leakage water which shall be piped to an approved discharge point.
 - (vii) Lubrication arrangements shall be designed to avoid any contamination of the pumped fluid.
 - (viii) Minimum lifespan of the cartridge type mechanical seals shall be 35,000 hours unless otherwise specified.
- (l) Bearings:
- (i) Pump bearings shall be external to pump casing and removable with the rotating element.
 - (ii) Pump bearing shall be adequately rated ball or roller type and shall be arranged to take up all radial and axial loads during start-up, shutdown and running conditions. Bearings shall be designed for loading 20 percent in excess of calculated maximum loading and shall be suitable for reverse rotation at 150 percent of rated speed or the maximum reverse speed that the pump can reach under installed conditions when driven backwards by reverse flow, whichever is greater. This requirement shall also be applicable for pumps with reverse rotation prevention mechanisms or devices.
 - (iii) Pump bearings shall be mounted in dust-proof housing and provided with accessible lubricating points for grease gun greasing.
 - (iv) Unless otherwise specified, the L-10 life of pump bearings for clean water pumps shall be at least 50,000 hrs. For used water (wastewater) pumping, this shall be at least 75,000 hrs unless otherwise specified.
 - (v) Bearing cooling arrangements if used shall be designed on the closed-circuit principle. Open discharge of cooling water into the pumping station drainage system shall not be permitted. The coolant flow shall be easily visible and local indication of bearing metal temperature shall be provided. Excessive metal temperatures shall result in automatic safe shut down of the pump.
 - (vi) Where water-lubricated bearings are used, means shall be provided to ensure that such bearings are supplied with adequate quantities of strained water at adequate pressure so that bearings are flooded with water before the pump starts and that a failure of the water supply results in safe shut down of the pump.

(m) Others:

- (i) The pump motor assemblies or pump and motor if installed separately shall be provided with jacking bolts to facilitate adjustments during installation.
- (ii) In the design and selection of the pump set, the following shall be taken into consideration to ensure that the pump set can operate satisfactorily, with minimal noise and vibration, and without cavitation or any other abnormalities:
 - 1) The various duty points as specified in Particular Specification;
 - 2) The level of pump impeller with respect to the water levels in the pump sumps;
 - 3) Other relevant site conditions e.g. presence of sewage in the pump sump; and
 - 4) The layout of the pumps and pipework configuration.
- (iii) The layout of the pumps and pipework configuration as shown on the Drawings is intended to be indicative and for the purpose of tendering only. The Contractor and the pump manufacturer shall confirm the adequacy and suitability of the pipework configuration, pump layout and structural slab elevations with respect to maximum length of intermediate shaft sections connecting motor and pump.
- (iv) The Contractor shall also confirm that the pump sets will operate satisfactorily without any hydraulic deficiency.
- (v) Any modifications or alterations to the layout for a proper and efficient operation of the pump supplied shall be the responsibility of the Contractor and shall be at no extra cost to the Contract.
- (vi) Any civil modifications for the proper installation of the new pump sets shall also be the responsibility of the Contractor and shall be at no extra cost to the Contract.
- (vii) The Contractor shall ensure the delivery of the pump manufacturer's supply of all relevant pipe clamps, lifting beams, ropes and slings for future overhauling of pump and motor.
- (viii) Where plug valves are used for pumps suction and discharge piping, these shall be motorised or pneumatically actuated regardless of the size and the local control (LCS) panel shall be installed on site near to the pump and subject to approval by the S.O..

G4.4.4 Pumping Auxiliaries

(a) Pressure Gauges:

- (i) Pressure gauges shall be installed on the pump delivery and pressure-vacuum compound gauge complying with the requirements of BS EN 837-1 shall be installed on pump suction.

- (ii) Locations of the suction and discharge pressure gauges shall take consideration of facilitating accurate determination of the pump head imparted to the pumped fluid by subtracting the suction pressure from the discharge pressure and compensating for the difference in the elevation between the suction and discharge measurement points, if any.
- (iii) All gauges shall be suitable for liquids or gases containing solids and corrosive medium. Diaphragm type shall be used for used water (wastewater) and sludge application unless otherwise specified.
- (iv) The normal operating pressure of the pump shall be approximately 60 per cent of the gauge range. Over range protection of 1.3 times the maximum scale reading shall be provided on the pressure gauges.
- (v) Gauge accuracy shall be within 2% of the full-scale range. All gauges shall be replaced if they do not maintain above the standard of accuracy for the duration of defect liability period.
- (vi) All pressure gauges shall be in accordance with BS EN 837 and have the following:
 - 1) A 100 mm diameter stainless steel case with a toughened window glass sealed and held in by a steel spring ring;
 - 2) A stainless steel movement mechanism filled with glycerine liquid to dampen the pointer from vibration & sudden impact;
 - 3) A diaphragm to suit the pumping media;
 - 4) A stainless steel connector of male NPT. Alternative thread type such as BSP shall be acceptable on the condition that the Contractor shall make the necessary provision for the connecting pipe to be tapped to the same threading type. The connection thread type for the isolation valve shall also be selected accordingly;
 - 5) A dial of black lettering and scales on a white background shall bear the legend showing service and units of graduation;
 - 6) A gauge cock to enable the gauge to be removed; and
 - 7) An isolation valves.
- (vii) Unless otherwise specified, the gauges supplied shall be of stainless steel construction and filled with pulsation dampening fluid. Additional pulsation dampers shall be installed wherever necessary.
- (viii) Stainless steel SS316 tubing, snubber and shut-off cock shall be provided and shall connect each gauge to its tapping. Gauges shall be flush mounted and labelled with their functions.
- (ix) Tappings shall be provided at both the suction and discharge flanges for connection of pressure gauges. For wastewater and sludge applications, pressure gauges shall be independently mounted on a stub branch line not less than 50 mm NB for pipe sizes of 50 mm and above. For pipe sizes below 50 mm the stub branch shall be the same size as the pipe. The tapping tubes shall be extended such that the suction and discharge pressures are measured at same elevation, unless otherwise specified or approved by S.O..

- (x) As far as possible, all pressure gauges shall be mounted either on walls, pumps supports or brackets that do not have any vibration. Direct mounting of the pressure gauges on to pump casings or pipes shall to be avoided. These shall also to be mounted at a height suitable for ease of reading. Where the pressure gauges are not visible from the operating floor level, they shall be grouped into a common panel. The location of the panel shall be agreed by the S.O. prior to installation. Suitable pipework, fittings and shut-off cock shall be provided to connect each gauge to its tapping. Gauges shall be flush mounted and labelled with their function.
- (b) Small-Bore Pipework and Fittings:
 - (i) The pump shall have tapped openings for air relief, drainage and gauge connections. All small piping associated with each pumping unit shall be of stainless steel SS316 pipe and nipples.
 - (ii) All tapped holes shall be fitted with stainless steel SS316 plugs. Suitable air relief valves, vacuum relief valves, combination air valves and strainers shall be provided as appropriate.
- (c) Pump Inlet and Discharge Pipework Manifold:
 - (i) Pump inlet and discharge pipework manifolds shall be pre-fabricated with the appropriate number and size of tapped bosses for the connection of pressure gauges, efficiency measurement equipment, vents, drains or injection points for chemicals, etc. Pipework manifolds shall not be drilled and tapped after installation.
 - (ii) A minimum of two sets of tapped bosses shall be provided on every pump inlet and discharge pipe. The tapped bosses shall be at least two pipe diameters from each pump flange face, preferably on a straight piece of pipe, and shall be on the horizontal or lower vertical centre line. Each boss fitted with ½ inch BSP or NPT female thread ball valves and plugged.
 - (iii) Air valves shall be connected to pipework via an isolation valve to allow the pipeline to remain in-service while maintenance of the air valve is carried out.
- (d) Flow Switch:
 - (i) Lubrication and cooling water pipelines to each pump shall be provided with a flow switch and a sight glass. The flow switch shall be of brass, incorporating an orifice, a drag disk and a micro switch and connectors, housed in an anodised cast aluminium enclosure. The flow switch shall be suitable for mounting in any position and shall be suitable for a maximum pressure rating of 16 bar.
 - (ii) The pump and the motor vibration monitoring shall consist of vibration sensors mounted on the pump and the motor if required. Refer to the relevant sections of the Electrical and ICA General Specification for more information.
 - (iii) Unless otherwise specified, dry running protection shall be provided.

- (e) Limit Switch:
 - (i) Minimum one (1) set of limit switches for the following:
 - 1) Manual suction valves to indicate the fully open and fully closed position. Where full open status is absent, the pump set shall not run; and
 - 2) Actuated delivery valve of each pump set to indicate the fully open and fully closed position and trip the motor if this valve is shut.
- (f) Filling / Flushing / Bleeding to prevent Pump Air-Lock:
 - (i) Vertically installed centrifugal pump especially for sludge application, shall be provided with filling/flushing/bleeding line to displace the gas/air generated/trapped in the pump casing. The filling/flushing/bleeding line shall consist of but not be limited to the following:
 - 1) Stainless steel pipework with non-return valve, isolation valve, automatic actuated filling/flushing/bleeding valve, pressure transmitter and associated fittings, electrical and instrumentation to connect water supply from plant water to the pump; and
 - 2) Automatic filling/flushing/bleeding of the pump prior to start-up to eliminate pump air-lock. The filling/flushing/bleeding shall continue and then stop when the required set pressure in the pump is reached.

G4.4.5 Source Quality Control

- (a) Refer to Pump Tests (Off Site and Factory Test) section.

G4.4.6 Design Requirements on Torsion and Vibration

- (a) Torsional and Lateral Vibration Analysis:
 - (i) The complete rotating assembly of motor rotor, pump shaft and impeller shall be designed such that the first critical speed of the assembly shall be not less than 1.4 times the maximum rotational speed of the pump (not less than $1.4 \times N_{\max}$ in revs/min).
 - (ii) The Contractor shall ensure that the pump manufacturer performs a complete lateral and torsional analysis of each distinct pump-motor system. This analysis shall identify the dry and wet lateral critical speeds plus the torsional critical speeds of the system. The excitation frequency range of the analysis shall include number of motor poles, and number of impeller vanes, including transient and steady state conditions. Appropriate lateral and torsional critical speed maps shall be produced, and no active critical speed shall be allowed within 20 percent of the operating speed range. This analysis shall be performed prior to fabrication of the equipment and is subject to review by the S.O..
- (b) Natural Frequency and Forced Vibration Analysis:
 - (i) Conduct natural frequency and forced response vibration analysis of the pump and connected piping, fittings, valves and maintenance platform at the pump operating speed.

- (ii) Natural frequency and all harmonics below 300 percent of maximum pump speed shall be calculated for the following structures:
 - 1) Suction piping and fittings from the concrete support block to the pump suction flange;
 - 2) Pump assembly and volute supports on concrete foundation of infinite stiffness;
 - 3) Discharge piping, fittings and valves, from the pump discharge flange through the concrete support block at the pump floor elevation; and
 - 4) Maintenance platforms and its access ladders.

Analyses shall demonstrate that piping and pump allowable working stress values are not exceeded due to vibration. Allowable piping working stresses corresponding to the design pressure, pipe wall thickness, diameter, operating temperature range and with the necessary allowance margin conforming with the guidelines of ASME B31.3 Process Piping shall not be exceeded.

- (c) Finite Element Analysis:
 - (i) Conduct finite element analysis of structural design of each pump and motor component.
- (d) The Contractor and the pump manufacturer shall co-ordinate the interfaces, designs and construction of the pump foundation, motor foundation and anchoring systems such that the above design requirements are complied with.

G4.4.7 Vibration Analysis and Monitoring System

- (a) Pump and motor and installations requiring vibration analysis studies shall fully conform to the requirements of this section.
- (b) The vibration monitoring system shall be able to measure the vibration levels in two (2) directions as a minimum, unless otherwise specified.
- (c) The vibration monitoring system shall be a complete package able to measure, monitor, record and output the required data and signals.
- (d) Unless otherwise specified in the Particular Specification, the following temperature sensors and vibration sensors shall be provided as minimum:

Motor per unit		
Temperature	6 nos.	2 nos. per phase , display (on LCP) 1 no. per phase
Vibration	4 nos.	2 nos. for motor drive end and 2 nos. for motor non-drive end. Inline (Discharge) & across direction on horizontal plane
Bearing temp	2 nos.	1 no. for motor end and 1 no. for motor non drive end
Intermediate shaft bearing per unit		(If applicable)
Vibration	2 nos.	1 no. per bearing at point (horizontal plane) where highest vibration.
Bearing temp	1 nos.	1 no. per bearing
Pump per unit		
Vibration	2 nos.	2 nos. for pump drive end and 2 nos. for pump non drive end. Inline (Discharge) & across direction on horizontal plane.
Bearing temp	2 nos.	1 no. pump drive end and 1 no. guild bearing
Pressure switch	2 nos.	1 no. for suction and 1 no. for discharge
Pressure transducer	2 nos.	1 no. for suction and 1 no. for discharge

- (e) A full list of the vibration and monitoring sensors for pumps and motor shall be submitted to S.O. for approval. The submission shall be expedited within 2 months of the Contractor's first submittal on pumps according to the approved project execution schedule to facilitate completeness check, selection as well as confirmation of the instruments and integration with the pumping equipment.
- (f) The vibration sensor shall incorporate accelerator probes unless otherwise specified.
- (g) Radial Shaft Vibration Transducers: Radial shaft displacement proximity probes shall be installed by the pump manufacturer. Orthogonal X-Y proximity probes shall be installed at each radial bearing housing. The probes shall contain a minimum 8 mm tip diameter mounted in a threaded stainless steel body. Each transducer shall consist of a probe and calibrated extension cable. All elements shall be fully protected in sealed weatherproof conduit and housings.
- (h) Thrust Position Transducers: Axial shaft displacement proximity probes shall be permanently installed by the pump manufacturer. Dual proximity probes shall be installed at each thrust bearing. The probes shall observe the thrust collar. The probes shall contain a minimum 8 mm tip diameter mounted in a threaded stainless steel body. Each transducer shall consist of a probe and calibrated extension cable. All elements shall be fully protected in sealed weatherproof conduit and housings.

- (i) Bearing Bonnet Velocity Transducers: Velocity transducer probes shall be permanently installed by the pump manufacturer. Transducers shall be located at the upper guide bearing. Two transducers shall be provided at 90 degrees between the two. The element shall be a velocity transducer with a velocity-to-displacement signal converter providing peak-to-peak displacement output. Minimum rated operating frequency shall be less than the minimum pump operating speed. Vibration element shall include shielded signal cable and shall be enclosed in an IP68 housing.
- (j) Bearing Temperature Elements: Each bearing shall be provided with two 100-ohm platinum resistance type temperature detectors (RTD) each of which shall be prewired to a terminal junction box mounted on the pump. The thrust bearing temperature elements shall be installed in two opposite thrust shoes of the thrust bearing. The metal of the bearings and mountings shall be drilled, milled, and tapped to place the element at the hottest points of the bearings and within 1.6 mm of the bearing metal.
- (k) The vibration and temperature transducers shall be installed in accordance with the guidelines provided by the American Petroleum Institute Standard 670, Fifth Edition for Vibration, Axial Position, and Bearing Temperature Monitoring Systems.
- (l) The bearing temperature and vibration transducers of pumps shall be utilised during all shop performance and running tests.

G4.4.8 Pump Performance

- (a) The Contractor shall guarantee the flow rate, the total head and the efficiency of the pump for all specified duty points when tested at the manufacturer's factory.
- (b) The pump shall operate at its duty point(s) within the acceptance tolerances for flow and total head as dictated in BS EN ISO 9906, Class Grade 1, unless otherwise specified. Submersible pumps shall follow different test grades depending on motor power rating as described under the section on submersible pumps.
- (a) Where pumps are working in parallel, curves for solo operation and all the various possible parallel combinations shall be submitted.
- (b) The pumps shall operate satisfactorily singly or in parallel for long periods without cavitation, deleterious effect, severe vibration or motor overloading.
- (c) Each pump shall be capable to run over the range of the minimum to the maximum static head under solo operation.
- (d) All positive displacement pumps shall be tested on accordance with BS EN 14343.
- (e) For critical pumps system, motors shall be designed to operate continuously. Motors shall comply with requirements in the relevant sections of the General Specifications.
- (f) In addition to confirming the hydraulic performance of the pump set, the test shall demonstrate that vibration is within the specified limits, the mechanical performance is satisfactory, the pump is free from overheating and noise level is within specified limited.

G4.4.9 End Suction Centrifugal Pump

- (a) Horizontally mounted pumps shall be mounted complete with drive motor on a common base plate. The pump/drive coupling shall be of the spacer type to facilitate removal of the pump rotating element and bearing housing without dismantling the pump casing, adjoining pipework or drive motor.
- (b) Vertically mounted end-suction pumps shall be supplied with suitable mounting stools. Motors shall be mounted on suitable brackets and connected to pump shafts with flexible couplings.
- (c) The shut off head shall be approximately 160% of the duty head.
- (d) Pump Construction:
 - (i) The dimensions of the pump shall be metric, conforming to BS EN 733 and BS EN 22858. The bedplate shall be of substantial fabricated steel construction with floor fixing bolt holes ready drilled.
 - (ii) The impeller shall be a one-piece casting. The impeller shall be completely machined on all exterior surfaces, except when high chrome alloy material is required. The interior water passages shall have uniform sections and smooth surfaces and shall be free from cracks and porosity.
 - (iii) For sludge and sewage application, impellers shall be provided with means to prevent abrasive matter reaching the glands. For fully shrouded impellers, means shall also be provided to prevent matter becoming trapped between the impeller vanes and the casing (i.e. enclosed non-clog impeller).
 - (iv) The impeller shall be securely locked to the shaft so that it will not be loosened by reverse rotation, but shall be easily removable.
 - (v) Pumps shall be fitted with suitable cartridge type mechanical seals. Flushing facilities shall be provided for mechanical seals where the fluid pumped may be contaminated with abrasive material. Provision for collection of seal leakage water shall be provided and shall be piped to an approved discharge point.
 - (vi) Lubrication arrangements for mechanical seal shall be designed so that there is no contamination of the pumped fluid.
 - (vii) As a minimum, the material of construction for the pump shall be as follows:

Frame	Cast iron, ASTM A48
Casing and Casing Covers	Iron; Cast iron, ASTM A48
Casing Wearing Ring	Martensitic stainless steel, Brinell 400+
Impeller	Iron, Cast Iron, ASTM A48
Impeller Wearing Ring	Martensitic stainless steel, Brinell 300 to 350
Shaft	Carbon steel AISI 1045
Shaft Sleeve	Stainless Steel AISI Type SS 316

Stuffing Box Hardware	Corrosion-resistant metal
Flexible Coupling	Resilient type
Bearings	Antifriction
Mechanical Seal	Durable type
Baseplate	Cast iron or fabricated steel

- (viii) Pumps shall be self-priming. If it is not possible, pumps and associated pipework shall be arranged, wherever possible, so that air can be completely expelled during priming. Where this is not possible, facilities shall be provided for the removal of the trapped air. Adequate facilities shall be provided for drainage of the pumps.

G4.4.10 Inline Centrifugal Pump

- (a) Installation of inline pumps shall be horizontal and the motor shall be on top of the pump. Any other arrangement of inline pumps is not acceptable.
- (b) Pumps shall be close-coupled, and fitted with single cartridge type mechanical shaft seals unless otherwise specified.
- (c) Casings shall have plugged provisions for suction and discharge pressure gauge connections as well as casing drainage. Means shall be provided for collection of seal leakage water, which shall be piped to an approved discharge point.
- (d) Pumps casings shall be provided with an integral baseplate for mounting on a well designed concrete foundation approved by the S.O., and a motor stool for mounting the drive motor. Motors shall comply with the electrical requirements in the relevant sections of the General Specification.
- (e) Lubrication arrangements shall be designed so that there is no contamination of the pumped fluid.
- (f) The minimum materials of construction for the pump shall be as follows:

Casing and Casing Covers	Iron; Cast iron, ASTM A48 Class 30
Impeller (for centrifugal pumps)	SS316 / Nodular iron GGG60 (BS Grade 600/3)
Shaft (for centrifugal pumps)	Stainless Steel 1.4021 (BS 420 S37)

G4.4.11 Double Entry Vertical/Horizontal Split Casing Pumps

- (a) General:
 - (i) Pumping units shall be designed for the operating conditions as indicated in the Particular Specification and drawings.
 - (ii) All specified conditions shall be at rated speed, unless otherwise indicated.
 - (iii) Overall (wire-to-water) efficiency for variable speed pumps shall include losses in the pump, motor, adjustable frequency drive, and any transformers supplied as part of the adjustable frequency drive equipment.

- (iv) Pump performance shall be stable and free from cavitation and noise throughout the specified operating head range at design suction submergences. The design performance shall be based on a wearing ring diametric clearance of not less than 25 μm per 25 mm of wearing ring diameter, or 300 μm total, whichever is greater.
- (b) Pump Construction:
 - (i) Vertical split case, single stage, double entry volute centrifugal pumps shall have the casing split along the axis of the shaft to permit removal of the impeller/shaft assembly without dismantling the pipework. Half casings shall be dowelled to ensure alignment. Jacking screws shall be provided on the casing flanges to facilitate dismantling. The bed plate shall be a substantial fabricated steel construction with floor fixing bolt holes readily drilled. The pumps shall be installed at the dry well with the spindle in a vertical position. The motor shall be located vertically above the pump and shall drive the pump by the flexible coupling and shafting. The motor and stool assembly shall be designed for ease of maintenance of the pump.
 - (ii) Horizontal split case, single stage, double entry volute centrifugal pumps shall have the upper and lower casing halves flange-bolted and doweled together with tapered dowels. The upper half casing flange shall have tapped holes for jacking screws and lifting lugs or eyebolts. Supporting feet, bearing arms, and nozzles shall be either cast integrally with or bolted and doweled to the lower half casing. Pipe tapped openings shall be provided for draining, priming and venting the casing as well as for draining stuffing box leakage. The pump feet shall be provided for bolting and doweled to a baseplate. The discharge volute shall be of the double or dual type to minimize hydraulic radial thrust.
 - (iii) The impeller shall be a one-piece casting completely machined on all exterior surfaces and dynamically balanced. The interior water passages shall have uniform sections and smooth surfaces and shall be free from cracks and porosity.
 - (iv) Impellers shall be of double suction enclosed type, manufactured from stainless steel type 316 unless otherwise specified. Impellers shall be of smooth finish and shall be statically and dynamically balanced on the spindle after assembly and keyed to the shaft.
 - (v) The impeller shall be keyed to the shaft and positively held in the centre of the discharge volute.
 - (vi) The shaft shall be completely machined. If the shaft is more than 60 mm in diameter, it shall be tapered at the coupling end. Deflection at the stuffing box shall not exceed 50 μm at any operating head.
 - (vii) The shaft shall be provided with sleeves extending from the impeller through each stuffing box. Each sleeve shall be positively secured to the shaft and shall be sealed to prevent leakage between the shaft and the sleeve. After assembly on the shaft, total runout shall not exceed 50 μm .
 - (viii) Renewable wearing rings shall be provided in the casing and on the impeller. The casing ring shall be positively locked against rotation in the lower half casing. The impeller ring shall be locked against rotation.

- (ix) When the rated head is greater than or equal to 70 m, the casing ring shall be shouldered for the full circumference of the ring.
- (x) The stuffing boxes shall each contain a single mechanical seal. The pumps seal shall not require an external source of seal water.
- (xi) Each unit and its drive assembly shall be supported on a single baseplate of neat design. Baseplates shall be provided with adequate openings to facilitate grouting and shall be designed to prevent the formation of trapped air packets during grouting. Provisions shall be available for alignment purposes such as adjustment screws.
- (xii) Fabricated steel equipment bases shall be hot dipped galvanized and finished coated or better. Bases may be machined after galvanizing.
- (xiii) Each pump shall be provided with lifting eyebolts or lugs, plugged gauge cock connections at the suction and discharge flanges, tapped and plugged openings for casing and bearing housing vents and drains, and appropriate fittings for adding bearing lubricant. Grease lubricated units shall be provided with a means of venting the casing.
- (c) Bearings:
 - (i) Inboard bearings shall be grease lubricated anti-friction type. The inboard bearing shall carry both radial and axial loads imposed by the pump.
 - (ii) The outboard bearing shall be a sleeve type bearing. The outboard bearing shall carry the radial loads imposed by the pump and drive unit.
 - (iii) All anti-friction bearings shall have an AFBMA L-10 Life Rating of 70,000 hours at specified operating conditions. The pump shaft speed shall not exceed the limits specified by the bearing manufacturer.
 - (iv) Anti-friction bearing housings shall be designed to maintain shaft alignment and ensure long bearing and lubricant life. Housings shall have labyrinth type running clearance designed to effectively retain the lubricant and keep out contaminants. Ample clearance for stuffing box maintenance shall be provided between the bearing housings and the stuffing box glands.
 - (v) All bearings shall be sealed to prevent contamination and fitted in machine housings located by dowels to assist maintenance. Pumps sets shall incorporate bearing re-greasing facilities and grease relief valves. Ball and roller bearings shall be steel caged.
 - (vi) If required as per manufacturer's recommendations, intermediate bearings shall be provided.
 - (vii) The thrust bearing shall be provided with a direct reading dial thermometer with a bulb or probe in a well, situated in the bearing metal to detect the highest temperature of the bearing. The thermometer shall have a 150 mm dial face and a stainless steel probe or capillary and bulb, and a range of 10° C to 150 °C.

- (d) Water Lubrication System:
 - (i) A bypass line from the pump discharge casing to the pump mechanical seals shall be provided if required for a completed functional system. The bypass line shall be of stainless steel tube, complete with stainless steel pipe, sockets and connectors not less than 25 mm in diameter. Each bypass line shall be provided with a sight glass and flow switch.
 - (ii) The Contractor shall submit the detailed pipe routing layout plans for the supply and discharge piping of the water lubrication system and ensure that obstruction, clash or trip hazards are avoided.
- (e) Coupling:
 - (i) The pump coupling shall be sized for continuous operation at full load and at maximum RPM when the misalignment is within the manufacturer's tolerance limit. Coupling design shall permit removal of the pump rotating element without disconnecting the piping, moving the drive unit, or causing axial movement of the coupling halves on the shafts. The coupling and shaft shall be enclosed with an approved metal guard.
 - (ii) The pump shall be directly connected to its driver by means of flexible coupling, vertical floating type.
- (f) Vibration Monitoring System:
 - (i) Split case pumps shall be provided with Vibration Detector system. The vibration detector system shall comprise vibration transducers fitted at each pump casings, motor casings, pump bearing drive ends and non-drive ends, motor bearing drive-ends and non-drive ends, intermediate bearings (if required by manufacturer's recommendations), vibration detector units, alarm delay units, an individual alarm unit and cabling.
 - (ii) The vibration transducer shall be piezoelectric accelerometer of the compression type and shall be sealed against dust and moisture. The vibration detector units shall be designed for continuous monitoring of vibration levels of the pumps and shall measure the vibration severity in accordance with BS ISO 20816-1.
 - (iii) The vibration detector unit shall have two (2) adjustable alarm levels:
 - 1) Level 1 – to give an alarm when maintenance is needed; and
 - 2) Level 2 – to indicate danger conditions and to trip the pumps.
 - (iv) The alarm delay unit shall be incorporated in the vibration detector system, to prevent the system during operating routines from responding to mechanical shocks which are not true representations of the vibration condition. The alarm delay unit shall be modular complete with light indications.

G4.4.12 Concentric (Tubular) Casing Pump / Vertical Turbine Pumps / Extended Drive Shaft Centrifugal Pump

- (a) General:
 - (i) Concentric (tubular) casing pump shall be of concentric casing with axial suction and tangential discharge.

- (ii) Vertical turbine pump shall be of the vertical pipe column and spindle, mixed flow or axial flow type.
- (iii) Extended drive shaft centrifugal pump shall be of the long shaft coupling and centrifugal flow type.
- (iv) The pump shall be provided with self-aligning type mechanical seal of an approved manufacturer for the seal housing.
- (v) Pump shall have stable characteristics such that the head falls continuously from a maximum at closed valve condition.
- (vi) Shaft lengths and suspension main lengths shall be determined for ease of installation and removal, bearing in mind the maximum head room available as shown on the Tender Drawings.
- (vii) The pump pipe column dimensions shall be so chosen as to avoid critical pump lengths and attendant vibration.
- (viii) The pump motors shall be located vertically above the motor floor level complete with direct coupled motor, face or flange mounted on top.
- (ix) The maximum speed of rotation shall not exceed 3000 rpm.
- (b) Pump Details:
 - (i) Pump casings shall be manufactured from spheroidal graphite cast iron in accordance with BS EN 1563 Grade 420/12.
 - (ii) Suction and discharge chambers shall be manufactured from fine-grained cast iron, and intermediate chambers shall be of the same material or stainless steel.
 - (iii) Impellers shall be manufactured from stainless steel Type SS 316 unless otherwise specified. Where appropriate or as dictated in the Particular Specifications, compatible corrosion resistant coating shall be applied. It shall be of smooth finish and perfectly balanced on the spindle after assembly to the shaft.
 - (iv) Impellers shall be fully shrouded and secured to the shaft by a key. The impeller retaining nut shall be fitted with a locking device. Shafts shall be stainless steel forgings compatible with the impellers.
 - (v) Wearing rings shall be of stainless steel Type SS 316 or ASTM A743 CF 8M or equivalent. All wearing parts shall be easily renewable.
 - (vi) The head plate shall be manufactured from fabricated steel. It shall be of heavy section and shall incorporate base flange for attaching to the foundation ring and thrust bearing. The base flange and thrust bearing support flange shall be machined true and provided with means to ensure correct alignment. The headpiece shall be of adequate strength to withstand the maximum hydraulic thrust that can be imposed upon it under closed valve conditions.
 - (vii) The head plate shall have large openings, adequate for maintaining the thrust bearing and the coupling.
 - (viii) The motor stool or mounting plate shall have a machined register on the topside to match a similar register which shall be provided on the base of the motor, so that the machine may be removed and replaced without disturbing the alignment.

- (ix) Foundation ring shall be manufactured from hard, close-grained cast iron and shall be machined true on all mating faces. It shall be of deep section so that it can be adequately grouted into the shaft cap. Holes shall be provided to enable air to be expelled during the grouting process.
- (x) Pump suspension main including elbow shall be manufactured from steel. All flanged faces shall be machined on faces and edges and spigotted to facilitate perfect alignment. All flanged faces shall be jointed with rubber O-ring or gasket. The interior and exterior of the suspension main shall be coated with non-toxic paint to a minimum dry film thickness of 75 microns.
- (xi) Intermediate steel shafting between pump and motor shall be provided with universal coupling at the bottom and top ends if required. The end couplings shall be designed to allow the pumps or motors to be removed without the need to dismantle the whole of the shafting. The coupling on the pump shaft shall be fitted on a sliding spline to allow vertical adjustment of not less than 25 mm.
- (xii) The drive shafts shall be dynamically balanced. The output shafts of both drive and driven equipment must be parallel within plus or minus one degree with offset or angular misalignment not to exceed 5 degrees.
- (xiii) The shafts shall be adequately sized to prevent whip or vibration when operating and shall be designed to accommodate small changes in alignment caused by foundation settlement, expansion and the like. Adequate shaft bearings and bearing supports shall be provided at intermediate floors.
- (xiv) All shafting shall be guarded up to the underside of the motor floor unless otherwise specified. Guards shall be demountable to facilitate shaft changes. Individual sections of guards shall not weigh more than 10 kg.
- (xv) Shafts and couplings shall be of machine ground stainless steel construction. Shafts shall be totally enclosed with shaft enclosing tube to prevent from contamination with the pumped water. All wearing surfaces at bearings and bushes shall be protected with removable sleeves in stainless steel. The couplings shall be taper bored, keyed and locked. The bores shall be perfectly true and concentric and the shafts and couplings shall be tested on assembly to demonstrate that they are true.
- (xvi) Shaft enclosing tube shall be of stainless steel and shall be designed with lifting lugs. All shaft enclosing tubes shall be jointed with rubber O-ring.
- (xvii) Shaft bearings shall be supported by flanged cast iron or Ni Resist spiders between the suspension main flanges. The bearings shall be protected from the pumped water by a tunnel tube into which clean water under pressure will be fed for bearing lubrication. The tunnel tubes shall be sealed at each bearing by Neoprene or EPDM (material to be corrosion resistant and compatible with pumped fluid) O rings and they shall be manufactured from stainless steel or non-corrodible material. No grease or oil lubricated bearings shall be accepted.

(c) Lubricating System:

- (i) Lubricating and cooling water to shaft bearings and mechanical seal shall be taken from the discharge branch main located at the plant and the water supply shall be tapped with stainless steel schedule 80 pipe from the service water piping main located in the vicinity of the pumping station. Duplex filters with differential pressure indication and control for automatic changeover shall be provided if required. The duplex filters shall also have visual indication for each filter element to alert the operator when blockage occurs.
- (ii) The Contractor shall submit the detailed pipe routing layout plans for the lubricating/cooling water and ensure that obstruction, clash or trip hazards are avoided.
- (iii) Where open loop lubricating and cooling water system is provided, lubricating and cooling water shall discharge to the plant drain system. Alternative discharge point shall be subjected to S.O.'s approval.
- (iv) Unless the water for lubricating the bearings is flowing properly, the switch attached to the lubricating system shall cause the motor to trip off or to prevent the motor from starting.
- (v) The Contractor shall supply and install a suitable lubrication and cooling water system designed to safeguard the bearings, thrust bearings and mechanical seals included in the Plant. The system shall be designed to avoid the use of booster pumps. The system shall be complete in every respect and shall include all necessary pipework, stop-valves, pressure reducing valves, pressure relief valves, solenoid valves, pressure gauges, visual flow indicators, strainers, meshed screens and other necessary equipment.
- (vi) The Contractor shall provide a temporal lubricating and cooling system for the pumps of concern during early phases of plant operation when permanent plant water may not yet be available. The temporal system may include but not necessary be limited to lubricating/cooling water storage tanks, the associated pumps, piping, valving, supports, instrumentation and control.

(d) Bearings:

- (i) The weight of the pump rotating parts and any hydraulic thrust shall be taken on a water-cooled thrust bearing which incorporates a journal bearing, or roller type thrust bearing. The bearing shall be of ample size to accommodate all possible loads plus an allowance of 20% and shall be suitable for reversed rotation. The bearing shall be provided above floor level and mounted in a separate housing under the motor.
- (ii) The vertical shaft bearings shall be designed for a minimum L 10 bearing life of 100,000 hours with special attention being taken in the selection of the proper tube size to avoid both the calculated critical speed and the half critical speed.
- (iii) Bearing spiders shall be drilled and tapped for a clean water feed to the bearings.
- (iv) A thermostat to detect high temperature shall be provided in thrust bearings and shall include contacts to initiate alarm in the respective starter board and to trip the pump when the temperature exceeds the predetermined value.

- (v) Delivery branches of each pump and delivery main pipe shall be fitted with gauges suitably scaled for maximum ranges of pump operation. All gauges shall be provided with isolating valve.

G4.4.13 Sump Dewatering Pumps

- (a) Sump dewatering pumps shall be of the vertical open impeller centrifugal type.
- (b) Pumps shall be supplied with all necessary discharge pipework, including check and isolating valves. Pumps shall be complete with suitable lifting gear for lowering and removing the pump from the sump. Pumps weighing 25 kg or more shall be lowered into the sump on guide rails and be connected to their respective discharge pipework with an angle flange and self-locating clamps. Pumps weighing less than 25 kg shall have the discharge nozzle connected by a short length of hose to the discharge pipework to facilitate ease of pump lowering into and removal from the sump. The hose shall be of the appropriate pressure rating and made of a suitable material.
- (c) Pump impellers shall be manufactured from cast iron and designed to pass solids of sizes up to 20 mm diameter unless otherwise specified and subject to approval of the S.O. with respect to the specific pumped fluid.
- (d) Pump casing shall be manufactured from cast iron unless specified.
- (e) Pump motors shall be of the submersible type. The depth of submergence shall be as specified or as required for the application.
- (f) The pumped medium shall not be used for cooling the pump motor.
- (g) Sump pumps of 1.5 kW and under shall incorporate an integral level detector, control and motor starter and shall be powered only with a suitably fused three phase or single phase low voltage supply and with supply isolation at the building distribution board.
- (h) Sump pumps over 1.5 kW shall be controlled and started from the building distribution board and be fed with a 3 phase supply. Control shall be via adjustable float level switches mounted adjacent to the pump.
- (i) Where appropriate, requirements of submersible pumps described in subsequent sections of this specification and of submersible motors described in the electrical sections of the General Specification shall apply.

G4.4.14 Vertical Dry-Pit Volute Pumps

- (a) Vertical dry-pit volute pumps shall be of the centrifugal or mixed flow type.
- (b) The pumps shall be suitable for continuous operation under the specified conditions and shall be capable of handling the pumped medium without undue wear and tear. The depth of submergence shall be as specified.
- (c) Where motor power ratings or/and raised installation elevations of motors are such that construction of submersible motors is not feasible or unnecessary respectively, non-submersible motors shall be acceptable. For such cases, it shall still be required for the pump volute casing to be submersible.
- (d) Pumps with motors mounted on elevated levels shall be driven with intermediate shafting segments.

- (e) Rotation of the pumps shall be clockwise viewed from the driver end. The direction of rotation shall be permanently marked in a clearly visible location.
- (f) Materials of construction other than specified in this general specification shall be according to the pump manufacturer's standard. AISI, ASTM, etc. references, types and grades specified are typical of material composition and quality. Equivalent materials will be considered.
- (g) Pump Construction:
 - (i) Major pump components shall be of the gray cast iron ASTM-A48-No 35B with smooth surfaces devoid of blow holes or other casting irregularities.
 - (ii) For applications in which the pumped fluid is sewage, pump volute casing shall be Ni-resist ductile cast iron and of non-concentric design with smooth passage for the specified solids size. Alternatively, grey cast iron, ASTM-A48 Class 35B material shall be guaranteed against corrosion as is required with Ni-resist ductile cast iron material.
 - (iii) The volute shall be provided with integrally cast or welded fabricated support feet for anchoring to sole plates on the foundation. The volute supports shall be designed to accommodate all static, dynamic, torsional and hydraulic loads occurring during pump operation. Provide matching steel sole plates and anchor bolts. Anchor bolts shall be provided as a shop fabricated assembly with a bolt locating frame to be embedded in the foundation concrete at the lower reinforcing steel mat. Furnish the assembly with a bolt locating template for holding the bolts during concrete placement. Sole plates shall include jack screws for levelling and grouting. Provide integrally cast or bolted steel plate lugs for lifting of the volute. The lowest point of the volute shall be fitted with a tapped drain boss, 75 mm minimum size.
 - (iv) The impeller shall be of corrosion resistant SS316L material. Alternatively, grey cast iron ASTM-A48 Class 35B shall be guaranteed against corrosion as is required with SS316L material. It shall be free of blow holes, projections, cracks and inclusions, dynamically balanced, multiple vaned, double shrouded non-clogging design having long throughlets without acute turns. The impeller shall be capable of handling solids, fibrous materials, heavy sludge and other matter found in the raw sewage. Mass moment of inertia calculations shall be provided by the pump manufacturer upon request. Impeller shall be keyed to the shaft, retained with an expansion ring and be capable of passing not less than the specified solid diameter.
 - (v) A wear ring shall be used to provide efficient sealing between the volute and suction inlet of the impeller. Each pump shall be equipped with a brass or Nitrile rubber coated steel ring insert that is drive fitted to the volute inlet. The pump shall also have an impeller wear ring made of stainless steel ASTM A320 Type 316, heat-shrink fitted onto the suction inlet of the impeller.
 - (vi) The pump suction and discharge nozzle shall be flanged and designed to the pressure rating taking account of the shut-off head of the pump and pump casing test pressure.

- (h) Means for prevention of impeller loosening and falling off (especially in situations of reverse flow such as during pump shut-down or trip) such as proprietary designed impeller locking mechanism, etc. shall be incorporated.
- (i) Means for prevention of motor reverse rotation by means of proprietary back-stop device, etc. shall be incorporated.
- (j) The pump shaft shall be an extension of the motor shaft. The pump shaft shall be of corrosion resistant SS316L material. Alternatively, ASTM S572 grade 50 material shall be guaranteed against corrosion as is required with SS316L material. The shaft shall not be in direct contact with the pumped liquid, designed to withstand impact and inertia forces arising from solids jammed between volute and impeller during operation at maximum speed. Calculations confirming shaft design shall be submitted.
- (k) All exposed bolts and nuts shall be made of Type 316 stainless steel to EN ISO 3506-1 or higher grade.
- (l) All metal surfaces in contact with the pumped media, other than the stainless steel, shall be protected by a factory applied water-based primer and finally covered by a two-component high-solid top coating.
- (m) Sealing design shall incorporate metal-to-metal contact between machined surfaces. Pump/motor unit matching surfaces where watertight sealing is required shall be machined and fitted with Nitrile or Viton rubber O-ring. Rectangular cross section rubber, paper or synthetic gaskets that require specific torque limits to achieve compression shall not be considered adequate or equivalent.
- (n) Each pump shall be tested and approved in accordance with National and International Standards (IEC 34-1, HI, CSA) and ISO 9906.
- (o) Pump motors shall have a built-in cooling system, which shall permit continuous operation at rated output both when in immersed condition and non-immersed condition.
- (p) Motor cable terminations shall be provided with a cable sleeve and strain relief.
- (q) Motor bearings shall be maintenance free.
- (r) The pump and motor shall be separated by two mechanical face seals. The lower seal shall rotate in the water medium and the upper seal shall rotate in an oil bath medium. Means of detecting seal leakage shall be provided and the grade and type of oil shall be subject to approval.
- (s) Where appropriate, requirements of submersible pumps described in subsequent sections of this specification and of submersible motors described in the electrical sections of the General Specification shall apply.
- (t) Cooling System Requirements shall be as follows:
 - (i) The pumped medium shall not be used for cooling the pump motor;
 - (ii) All required flows and pressures for the cooling media at points of feed to the pump and motor shall be clearly specified;
 - (iii) An integral, self-supplying cooling system for each pump shall be provided. Where large capacity of pump necessitates external cooling system, this shall be clearly made known;

- (iv) It shall be noted plant utility water may not be available as the source of cooling media supply during testing and commissioning stages and early stages of plant operation. As such, the Contractor and the pump supplier shall incorporate the alternative supply of cooling media for the pumps and motors requiring it; and
- (v) The water jacket for integral, self-supplying cooling system of the pump motor shall encircle the stator housing and shall be made of cast iron ASTM A48-No35B. The jacket shall thus provide heat dissipation for the motor located in the dry pit and surrounded by air. After passing through a classifying labyrinth, the impeller back vanes shall provide the necessary circulation of the cooling liquid which is a portion of the pumped media through the cooling system. An air evacuation tube shall be provided to facilitate air removal from within the jacket. Drilled and threaded provisions as additional means for external cooling, seal flushing and air relief shall be provided for completely non-submerged pump operation in air having temperature of up to 40 deg. C.
- (u) Bearing requirements shall be as follows:
 - (i) Grease lubricated bearings shall be provided for the shafts;
 - (ii) The upper bearing for taking radial forces shall be a single roller bearing;
 - (iii) The lower bearing shall consist of at least one roller bearing for radial forces and one angular contact ball bearing for axial thrust. The lower bearing housing shall include an independent thermal sensor to monitor the bearing temperature as specified. If a high temperate occurs, the sensor shall activate an alarm and shut the pump down;
 - (iv) Statistical design life at the 10 percent failure level shall be not less than 100,000 hours;
 - (v) Each bearing shall be provided with PT 100 duplex type bearing RTD for temperature monitoring and shall be connected to the Motor Protection Relay of the respective pump-motor; and
 - (vi) Each bearing shall be provided with vibration sensors which shall be connected to the Condition Monitoring System of the respective pump-motor.
- (v) Mechanical Seal requirements shall be as follows:
 - (i) The mechanical seal shall be of the totally split with clean water flushing type. It shall also be of the automatic aligning type with independent self-aligning rotating and stationary parts;
 - (ii) The wear surfaces of the seal shall be removable from the seal for inspection or replacement without any special tool;
 - (iii) The seal shall be designed such that it can be dismantled for maintenance without removing pump or motor;
 - (iv) All metal parts of the seal shall be Type 316 stainless steel. The material for the rotating face and stationary face shall be silicon or approved equivalent. The material for the 'O' ring shall be fluorocarbon or fluoric rubber or approved equivalent suitable for use in raw sewage;

- (v) The seals shall require neither maintenance nor adjustment and shall be capable of operating in either clockwise or counter clockwise direction of rotation without damage or loss of seal function; and
- (vi) Should seals fail and allow fluid to enter the stator housing, the fluid shall not come into contact with the lower bearing or the stator.
- (w) Intermediate Drive Shaft requirements shall be as follows:
 - (i) Universal joint type intermediate shafting shall be provided to transmit power between the motor and pump;
 - (ii) The intermediate shafting shall consist of the number of sections necessary to span the distance between the motor and pump;
 - (iii) The shaft shall have a telescoping splined slip member which will assist in removal of the pump shaft and shall accommodate thermal expansion and structural floor deflection up to 75 mm;
 - (iv) Intermediate drive shaft design shall provide for support of the shaft in place, without the requirement for complete shaft removal when the motor is uncoupled for maintenance;
 - (v) The pump manufacturer shall make known any requirement for a small shaft tilt angle in the range of 0.5 to 3 degrees to facilitate more effective lubrication of the bearings and load the bearings to prevent looseness;
 - (vi) The pump manufacturer shall make know the maximum allowable shaft deflection which shall be measured following installation and not be exceeded;
 - (vii) Universal joint bearings shall provide a minimum of 50,000 hours B-10 life, sized at maximum power and maximum rpm;
 - (viii) Universal joint bearings shall be designed with industrial roller bearings. Needle bearings will not be accepted;
 - (ix) Universal joints shall be housed in the mating flange yokes and weld yokes by a split bearing ear design to allow ease of replacement;
 - (x) Each bearing cap shall be equipped with a lubrication fitting and a bronze protection dome;
 - (xi) All bearing greasing shall be performed by one or more automatic greasing system(s) designed to the optimum configuration and consisting of grease dispensers able to lubricate all bearings at prescribed timings recommended by the manufacturer;
 - (xii) Shafting shall be high strength low alloy metallic DOM (drawn over mandrel) tubing;
 - (xiii) Shaft and universal joint assemblies shall be dynamically balanced at maximum pump rpm to 7.9×10^{-5} kg-m per kg of shaft weight per end; and
 - (xiv) Safety guards shall be provided and supported by the bearing bonnet as required by Occupational Safety and Health Administration (OSHA) and local statutory regulations. Guards shall be hot dip galvanised and shall include hinged doors at lubrication points.

- (x) Access Work Platform requirements shall be as follows:
 - (i) Provide each pump with an access work platform, safety rail and ladder for the inspection and maintenance of the pump itself, shaft coupling, bearings and seal. The platform shall be able to access the motor to cater to all maintenance associated with the motor;
 - (ii) The access platform shall be fabricated of steel structural shapes and plates, and shall be hot dip galvanised after fabrication and finishing;
 - (iii) Platform loading design shall be based on 140 kg concentrated load on any part of the platform and ladder;
 - (iv) The platform and ladder shall be easily removable for pump maintenance; and
 - (v) Hardware and fasteners shall be Type 316 stainless steel.
- (y) Pump Components Machining requirements shall be as follows:
 - (i) Machining of all pump components, other than proprietary outsourced products such as bearings and seals, shall be done in the pump manufacturer's shop facilities.
- (z) Pump Balancing requirements shall be as follows:
 - (i) Balancing of the impeller, complete with wearing rings, shall be by removal of metal by light eccentric surface machining of the external surfaces of the impeller shroud(s);
 - (ii) The area machined for balancing purposes shall be smoothly transitioned such that changes in shroud thickness shall be gradual in all directions;
 - (iii) Drilling of holes or addition of weights for balancing purposes will not be accepted;
 - (iv) The rotating assembly of the pump/motor shall be statically and dynamically balanced;
 - (v) Balancing shall be by removal of metal from a special balancing ring attached to the pump shaft;
 - (vi) Removal of metal by drilling of rotating components other than the balancing ring will not be accepted;
 - (vii) The Contractor shall submit detailed procedures for balancing of the impeller and the rotating assembly; and
 - (viii) Balancing certificates shall be submitted for all impellers and rotating assemblies, including spares.

G4.4.15 Submersible Pumps

- (a) Submersible pumps shall be as specified and for the required operating conditions. The impeller shall be of mixed flow type unless otherwise specified.
- (b) Where applicable, submersible pumps shall be capable of handling a mixture of water, sand, sludge and other small solid particles.
- (c) Pumps shall be suitable for continuous operation under the specified conditions in handling the pumped medium without undue wear and tear. The depth of submergence shall be as specified.

- (d) When required, pumping units shall be designed so that maximum reverse rotation due to reverse flow at the head as required will not cause damage to any component. The Contractor shall ensure that the pump supplier coordinates this provision with the motor manufacturer.
- (e) Pumps shall be capable of allowing free passage of solids up to the size as specified. The impeller shall be machined to tight tolerance limit and shall be dynamically balanced.
- (f) The cylindrical shroud shall be suitably designed to withstand the stress due to hydraulic pressure generated by pump set as well as water surge that may arise during pump stop. Sufficient clearance shall be provided between the shroud and pump set to facilitate heat removal and prevent formation of stagnant fluid. The shroud shall be designed to avoid cavitation and high pumping noise.
- (g) The pump casing shall be of high quality grey cast iron to BS EN 1561, impeller of stainless steel to BS EN 10293 Grade SS 316 C16 and the shaft shall be of stainless steel to BS EN 10095, BS EN 10250-4, BS EN 10085, PD 970, BS EN ISO 683-4, BS EN ISO 683-1, BS EN ISO 683-2 and BS EN ISO 683-3 Grade 431 S29. The shaft sleeve shall be of leaded gunmetal to BS EN 1982.
- (h) The impeller casing shall have well-rounded water passages and smooth interior surfaces free from cracks, porosity, blowholes, or other irregularities. The discharge nozzle shall be flanged and sufficiently rigid to support the pumping unit under all operating conditions.
- (i) The impeller shall be a semi-open one-piece casting with not more than two non-clog passages. The interior water passages shall have uniform sections and smooth surfaces and shall be free from cracks and porosity.
- (j) For pumping units 15 kW and larger, replaceable wearing rings shall be provided in the casing and on the impeller. The rings shall be positively locked in place.
- (k) For pumping units less than 15 kW a replaceable wearing ring or axially adjustable wearing plate shall be provided in the casing. Casing wearing ring shall be securely fastened to the impeller casing front cover to provide either an axial or radial running clearance. Axially adjustable wearing plate shall be arranged to permit adjustment of the axial running clearance between the impeller and plate. The wearing plate shall have an outward spiralling groove designed to force stringy solids outward and away from the impeller.
- (l) The oil chamber shall contain a drain plug and a vent plug.
- (m) Each pump shall be provided with two mechanical rotating shaft seals arranged in tandem and running in an oil chamber. Each interface shall be held in contact by an independent spring system designed to withstand maximum suction submergence. The seals shall require neither maintenance nor adjustment and shall be readily accessible for inspection and replacement.
- (n) Shaft seals lacking positively driven rotating members or conventional double mechanical seals which utilise a common single or double spring acting between the upper and lower units and requiring a pressure differential to offset external pressure and effect sealing will not be acceptable. The seals shall not rely upon the pumped media for lubrication and shall not be damaged if the pumps are run un-submerged for extended periods while pumping under load.

- (o) All mating surfaces of major components shall be machined and fitted with O-rings where watertight sealing is needed. Sealing shall be accomplished by O-ring contact on four surfaces and O-ring compression in two planes, without reliance on a specific fastener torque or tension to obtain a watertight joint. The use of elliptical O-rings, gaskets, or seals requiring a specific fastener torque value to obtain and maintain compression and water-tightness will not be acceptable. The use of secondary sealing compounds, gasket cement, grease or other devices to obtain watertight joints will not be acceptable.
- (p) Pumps shall have stable characteristics and be capable of continuous operation at within $\pm 50\%$ of the design flowrate.
- (q) Pump motors shall have a built-in cooling system, which shall permit continuous operation at the rated output.
- (r) The pumped medium shall not be used for cooling the pump motor.
- (s) For pump motor ratings less than 40 kW, pump tests shall conform to BS EN ISO 9906: 2012 Grade 2, unless otherwise specified or requested by S.O.. Grade 1 test shall apply for pump with motor ratings of 40 kW and greater.
- (t) Motor cable terminations shall be provided with a cable sleeve and strain relief.
- (u) The insulation rating of the motor shall be Class F rated to run at Class B. The motor ingress protection index shall be IP68 to BS EN 60529.
- (v) Motor bearings shall be lubricated for life and maintenance free. These bearings shall have metric dimensions and conform to BS ISO 10317.
- (w) The pump and motor shall be separated by two mechanical face seals. The lower seal shall rotate in the water medium and the upper seal shall rotate in an oil bath medium. The lower seal shall prevent the pumping fluid from entering the oil chamber and the upper one shall stop the oil from entering the motor compartment. The sealing rings of the mechanical seals shall be of tungsten carbide or other better materials suitable for the rotating speed and system pressure. Means of detecting seal leakage shall be provided. The grade and type of oil shall be subject to approval.
- (x) Moisture sensor shall be provided.
- (y) Submersible motor requirements shall be according to the electrical sections of the General Specification.
- (z) Lifting Facility for submersible pumps shall include, but not be limited to the following:
 - (i) The guide rail lifting system shall include guide rails, pump set pedestal discharge elbow, discharge base, holding down bolts, lifting chain and all mounting brackets. Where necessary, pump base skid made of corrosion resistant material shall be provided to ensure the horizontality of the base.
 - (ii) The base shall be sufficiently rigid to firmly support the guide rails, discharge piping, valves and pumping unit under all operating conditions. The base shall be provided with one or more integral support legs or pads suitable for bolting to the floor of the wet well.

- (iii) The face of the discharge elbow inlet flange shall be perpendicular to the floor and the mating part of the pedestal discharge elbow and the discharge flange of the pump shall truly align and form a perfect match without the requirement of bolting.
- (iv) Sealing at the discharge connection shall be accomplished by metal to metal contact of machined faces or a suitable gasket. Leakage of pumping medium from the discharge connection during operation of the pump shall not be accepted.
- (v) The lifting attachment of the pump set shall be designed such that by lowering of the pump set its discharge flange shall self-click onto the connecting flange of the pedestal discharge elbow. The pump and motor assembly shall be automatically connected to and supported by the discharge base and guide rails so that the unit can be removed from the wet well and replaced without the need for operating personnel to enter the wet well. When the pump is lifted, it shall be able to detach from the pedestal discharge elbow with ease.
- (vi) The weight of the pump set shall be taken up by the pedestal. No portion of the pump set shall rest on the floor.
- (vii) Each guiderail mounted pumping unit shall be provided with an integral, self-aligning guide rail sliding bracket. The bracket shall be designed to obtain a wedging action between flange faces as final alignment of the pump occurs in the connected position. The bracket shall maintain proper contact and a suitably sealed connection between flanged faces under all operating conditions. The sliding bracket shall be non-sparking.
- (viii) Each guide rail mounted pumping unit shall be equipped with one or more guiderails. Guiderails shall be sized to fit the discharge base and the sliding bracket and shall extend upwards from the discharge base to the pump access opening. An upper guiderail bracket shall be provided at the pump access opening.
- (ix) Each guide rail mounted pumping unit shall be provided with a lifting chain suitable for lifting and lowering the pump. Lifting chain shall be attached to the top of the wet well.
- (x) Lifting chain shall have safe working load with adequate safety factor which commensurate with the weight of the pump set. This load capacity rating shall account for the greatest load situation with resistance arising from friction of the guide rails due to the fluid and articles inherent in the fluid. The lifting chain shall comprise closed lifting rings at regular intervals and its ends. The lifting ring at the lower end shall attach to the lifting eye or handle of the pump by shackle(s) whereas the upper ring shall normally be hung onto the guide rail mounting bracket. Certificate of test and thorough examination of the lifting facilities in compliance with statutory requirements shall be furnished with the equipment. Design of the lifting chain shall eliminate all possibilities of failure.
- (xi) The design of the lifting chain, guide rails, hold down bolts and nuts shall be by the manufacturer. Sizing calculation of the holding down bolts and installation drawing of the pedestal discharge elbow and guide rail lifting system shall be submitted.

- (xii) Each skid mounted base shall be provided with a fabricated steel skid extending from the bottom of the volute to the floor for the purpose of supporting the pump as a free-standing unit.
- (xiii) The lifting chain, guide rails and hold down bolts and nut shall be of type 316L stainless steel to BS EN 10095, BS EN 10250-4, BS EN 10085, PD 970, BS EN ISO 683-4, BS EN ISO 683-1, BS EN ISO 683-2 and BS EN ISO 683-3 Grade SS 316. For more severe corrosive applications, materials of construction such as duplex stainless steel or super duplex stainless steel shall be used. Such applications are elaborated in the Particular Specification.
- (xiv) Hold down bolts and nuts shall be accurately located and shall be P.E. endorsed for the design.
- (xv) All iron and steel parts which shall be in contact with pumped liquid or submerged after installation, including the inside of the casing, the impeller, and the discharge elbow, shall be shop cleaned in accordance with the coating manufacturer's recommendations and painted with the epoxy coating system specified.
- (xvi) All other iron and steel surfaces, except stainless steel and machined surfaces, shall be protected with suitable protective coatings applied in the shop.
- (xvii) Electric Motors shall comply with the following as a minimum:
 - 1) Each pump shall be driven by an air-filled, totally submersible electric motor provided by the pump manufacturer. Motor nameplate rating shall exceed the maximum power required by the pump in the operating head range. Each motor shall have a voltage, frequency and phase rating as required and shall have a service factor of 1.5. The stator housing shall be an air-filled, water tight casing. A cooling jacket shall encase the motor housing for each pump where needed to maintain adequate cooling. Cooling jacket shall require no external source of cooling water unless otherwise indicated. Motor insulation shall be moisture resistant, Class F, 155 ° C. Each motor shall be NEMA Design B for continuous duty at 40 °C ambient temperature, and designed for at least 10 starts per hour.
 - 2) The motor bearings shall be anti-friction, permanently lubricated type. The lower bearing shall be fixed to carry the pump thrust and the upper bearing free to move axially. The bearings shall have a calculated AFBMA L10 Life Rating of 50,000 hours when operating at maximum operating head. Maximum shaft runout at the mechanical seals shall not exceed 50 µm at any point in the operating head range.
 - 3) Each motor installed in a wet well shall be capable of continuous operation in air (unsubmerged) for at least 24 hours under pump full load conditions, without exceeding the temperature rise limits for the motor insulation system.

- 4) All cables for wet well mounted pumps shall terminate in a junction box outside the wet well. Cables shall have sufficient length of slack which will be coiled on a cable hook at the top of the wet well. Each cable shall be supported by AISI Series 300 corrosion-resistant stainless steel or woven grips to prevent damage to the cable insulation. Better materials shall be used where necessary to match higher grade materials specified for pump components and accessories in more corrosive environments as elaborated in the Particular Specification.
- 5) The cable entry water seal shall include a strain relief and a grommet type seal designed so that a specific fastener torque is not required to ensure a watertight, submersible seal. The cable entry junction box and motor shall be separated by a stator lead sealing gland or a terminal board. The junction box shall isolate the motor interior from moisture gaining access through the top of the stator housing.

G4.4.16 Double Disc Centrifugal Pump (For Sludge and/or Scum Pumping Application only)

- (a) Double disc centrifugal pumps are meant for pumping either scum and Nocardia foam or drain water that could contain long fibrous material and solid particles only.
- (b) The pump shall comply with the requirements of the submersible or centrifugal pump specification with the following additional requirements:
 - (i) The pump shall be single passage and clog free with the ability to handle rags or other fibrous material without plugging;
 - (ii) The impeller shall be double disc type and able to handle minimally 15% to 30% air entrainment when pumping at low level position;
 - (iii) The impeller design shall permit low liquid velocities and gradual acceleration and change of flow direction of the pumped media (Boundary layer viscous drags principle). Impeller/casing design shall result in a single helical passage free of surfaces to which solid or fibrous material can adhere;
 - (iv) The impellers shall be made of stainless steel 316L or equivalent or better material, unless specified otherwise;
 - (v) The pump casing shall be of cast iron ASTM A48 Class 30, unless otherwise specified. It shall have sufficient strength, weight and metal thickness to ensure long life, accurate alignment and reliable operation;
 - (vi) The liner material for the suction and the internal lining of casing of the pump shall be abrasion resistant Ni-hard with a minimum Brinell hardness of 350 or better;
 - (vii) The bearing housings shall be made of Cast Iron, ASTM A48CL30 or equivalent or better. Bearings shall be of the ball or roller type designed for a minimum L10 bearing life of 50,000 hours at best efficiency point. Bearing shall be grease lubricated and automatic grease lubrication system shall be provided;
 - (viii) The material of the shaft shall be Stainless Steel 316 as minimum and shall have sufficient strength to transmit full driver output;

- (ix) Unless otherwise specified, the mechanical seal shall be provided for submersible double disc centrifugal pump type; and
- (x) The motor shall be suitable for dry well installation application. For this type of application, the pump and the pumped medium shall not be used for cooling the pump motor.

G4.4.17 Chemical Dosing Pumps

(a) General

- (i) Specific requirements for chemical dosing pumps are described in the respective sections of the Particular Specification.
- (ii) Chemical dosing pumps shall be either reciprocating diaphragm type or air-operated double diaphragm type or Peristaltic type unless otherwise specified.
- (iii) Dosing pumps shall be mounted in a bund at an elevation above the bund wall and fully meet all operating requirements unless otherwise specified.
- (iv) Pumps shall be provided complete with suction and delivery isolation valves, check valves, pressure gauges, pressure relief valves discharging to a catchpot (one per chemical) fitted with a level switch for alarm initiation and pump trip, pulsation dampeners, loading valves, suction strainers, calibration columns and dosing lines terminating in injection tubes supplied with non-return and isolating valves.
- (v) Dosing pumps shall be fully accessible for operating and maintenance purposes without personnel having to enter the bund itself.
- (vi) Where the pumps are mounted in the tank bund, the pumps shall be mounted on a plinth with stanchion or framework supports where necessary and above the flood level in the bund. All materials of construction for the supports shall be corrosion resistant.
- (vii) Pumps shall be driven by a flange mounted IP 65 motor as minimum.
- (viii) Suction and discharge valve shall be the single ball type allowing a free flow self-cleaning action. Ball and seat materials shall be resistant to abrasion and compatible with the chemical being handle.
- (ix) The pump design and materials shall be entirely compatible with the chemical to be pumped with special regard to the following chemical properties:
 - Corrosive nature;
 - Concentration;
 - Temperature;
 - Viscosity;
 - Specific gravity;
 - Abrasiveness;
 - Ability to generate free gas; and
 - Volatility.

- (x) Automatic air venting system shall be provided for pumping liquids containing gases or with tendency for gassing. The system shall include solenoid or motorised valve in the discharge pipe close to the pump head and vent it back to the supply tank at high level. The solenoid or motorise valve shall operate a few minutes to vent and purge accumulated gas in the discharge line when the pump initially starts before switching back to normal pumping system. The automatic air venting system shall be submitted and approved by the S.O.. All materials shall be compatible with the liquids being pumped. Spring loaded discharge check valve in this application shall not be accepted unless approved by S.O..
 - (xi) Pump inlets and outlets shall be fitted with valve connections to permit flow calibration during pump commissioning. A calibration tube shall be installed with each pump.
 - (xii) Dosing pumps shall be provided with facilities for flushing and drains. A group of duty and standby dosing pumps shall be provided with one calibration tube on the pump suction.
 - (xiii) Pumps shall be self-priming and shall meet the performance requirements as specified.
 - (xiv) The pumps shall be provided with a diaphragm type pressure gauge on the delivery side of the pump. The gauge shall be mounted on the delivery pipework.
 - (xv) In addition to the over pressure sensor, an adjustable pressure relief valve shall be provided in pressure relief pipework, discharging to a catch pot fitted with a level probe unless specified otherwise.
 - (xvi) The pressure relief valve shall be of the spring-loaded type with the spring external to the valve. It shall be possible to set the operating range within 50 to 300% above the normal working pressure. The actual setting pressure shall be above the setting of the over pressure switch but less than the maximum pressure the pump can develop.
 - (xvii) The diameter of the pressure relief valve and pipework shall be the same as the pump delivery main unless otherwise specified. No other valves shall be fitted to the relief lines.
- (b) Pump Performance:
- (i) All units of dosing pumps for any particular application shall be sized for the required capacity covering the full operating range of dosing rates of the specific chemical by means of turn-down ratio. The capacity shall be delivered at the head required to drive the chemical from the chemical storage tanks to the designated points of chemical injection. If special features such as independently powered fans are required to achieve the specified turndown, these shall be included.

- (ii) The performance characteristics of the dosing pumps shall be adequate in terms of linearity, accuracy, reproducibility and reliability to achieve the stated plant performance guarantee. The deviation from flow linearity of the pump shall not exceed $\pm 3\%$ of the rated capacity. The steady state accuracy shall not exceed $\pm 1\%$ of the mean turn down range. The flow rate repeatability expressed as a percent of the rated capacity of the pump shall not exceed $\pm 3\%$ of the rated capacity. Where no guarantee is stated, then the pumps shall at least comply with API Standard 675, Positive Displacement Pumps — Controlled Volume, Section 2.1.
- (c) Pump Control:
 - (i) Drive motors shall be of the three-phase cage induction type whether for fixed speed or variable speed operation.
 - (ii) Pumps shall incorporate a variable stroke mechanism to allow the output to be varied while the pump is running. The stroke adjustment mechanism shall have the facility to be operated either manually by a control knob or automatically by a motorised actuator. The variable stroke mechanism shall give stepless adjustment between zero and maximum stroke length. A stroke length indicator shall be provided.
 - (iii) All variable speed drives and electrical stroke positioners shall be provided with speed/position control and feedback to/from the SCADA system unless otherwise specified. Where variable speed operation is specified, the speed turn-down ratio shall be not less than 5:1.
 - (iv) A stroke positioner and flow proportional/compound loop controller shall be provided for automatic stroke length adjustment and shall include the following as minimum:
 - 1) A reversible motor;
 - 2) A feedback potentiometer;
 - 3) Manual override, with indication that the stroke positioner motor has been disengaged; and
 - 4) 3 alarm contacts with indication shall be provided as minimum:
 - Stroke – High;
 - Stroke – Low; and
 - Stroke Positioner Motor – Disengaged.
 - (v) The pump set shall be capable of providing the operating turndown range, as determined by the process design requirements either by means of engaging stroke length adjustment or stroke frequency adjustment or a combination of both.
 - (vi) When flow proportional dosing is required, the variation of output shall be achieved by varying the speed of the pump motor and not the pump stroke length.

- (vii) Where automatic stroke control alone is able to deliver the required full operating ranges of output flow and corresponding head for specific applications of chemical dosing pumps, these pumps shall be provided without VFD control. Otherwise, all chemical dosing pumps shall be provided with both stroke length and stroke frequency adjustment capabilities.
- (d) Dosing Pump Calibration Tube:
 - (i) A valved branch on the suction side of each group of duty and standby pumps and a calibrated transparent tube in a support frame shall be provided to facilitate checking of the volumetric capacity of each pump in situ.
 - (ii) Calibration tube shall be transparent, manufactured of material resistant to the chemical and not be discoloured by the chemical.
 - (iii) The effective height of the tube shall not be less than 1.25 m and the diameter shall be selected to ensure that at the maximum pumping capacity the rate of fall of level in the tube shall not be greater than 1.0 m/min. The height of the tube shall be such that when full, the liquid level in the calibration tube shall be the same as that in the stock tank. This may be achieved by providing the calibration tube with a vent pipe.
 - (iv) The calibration tube shall be provided with a drain discharging to an appropriate bunded area.
- (e) Y-Strainer:
 - (i) Strainer shall be installed in the inlet piping with manufacturer's recommendations incorporated where applicable. The strainer shall have a transparent Y-pattern body with 35-mesh screen and O-ring seals. The material of the strainers shall be resistant to the chemical.
- (f) Pulsation Dampener:
 - (i) Diaphragm type pulsation dampeners shall be fitted where necessary to the suction and/or discharge side of dosing pumps.
 - (ii) Each pulsation dampener shall be fabricated of materials resistant to the pumped liquid and consistent with the pipe materials, and shall include an air charging valve and air gauge.
 - (iii) The pulsation dampener shall be sized to limit pressure surges to between 5 and 10 percent of the mean pressure.
 - (iv) The dampeners shall be of the vertical air vessel type connected in a manner which minimises the possibility of blockage occurring.
 - (v) An air supply shall be connected to the dampeners to facilitate periodic re-charging. The supply shall incorporate a filter, pressure regulator and pressure gauge.
- (g) Pressure Gauge and High Pressure Switch:
 - (i) Pressure gauge and high pressure switch with isolating diaphragm shall be installed in the discharge piping.

- (ii) The high pressure switch shall be of the diaphragm type, shall have a weatherproof housing, shall have a trip point reliability better than 1 percent of actual pressure, and shall at minimum be SPDT rated at 24 volts DC. The isolating diaphragm shall protect the pressure gauge from the chemical solution.
- (h) Pressure Relief Valves:
 - (i) Pressure relief valves shall be fitted to the delivery side of dosing pumps and shall discharge in a safe manner to a catch pot with level sensor designed for this purpose.
 - (ii) The valves shall be of a non-choking type free of cavities and crevices which are likely to be blocked.
 - (iii) The valves shall discharge back to the suction side of the pump.
 - (iv) Provision shall be made for operating the valves manually for purposes of flushing.
 - (v) Loading Valves.
 - (vi) Loading or back-pressure valves shall be fitted where necessary to the delivery side of the dosing pumps. All materials of construction shall be fully compatible with the chemical handled.
- (i) Air-Operated Double Diaphragm Pumps:
 - (i) For Air operated double diaphragm pumps specified for dosing applications, these shall meet the following additional requirements as minimum:
 - 1) Pumps shall be of the positive displacement, double diaphragm and non-lubricating type containing two chambers wherein the pumping action is generated by hardwearing, flexible, synthetic diaphragms connected by a rod oscillating within the two pumping chambers containing the pumped liquid.
 - 2) Ball type non-return valves shall be used on the inlet and outlet to each chamber. Materials of wetted parts shall be compatible with the liquid being pumped. Balls shall be of steel with wear resistant elastomer lining or neoprene or Viton or stainless steel or ceramic. Valve seats shall be stainless steel grade 316S16 to BS EN 10213. Diaphragm shall be neoprene or Viton. Diaphragm rod shall be stainless steel grade 316S16 to PD 970.
 - 3) Diaphragm movement shall be provided by compressed air directed to each chamber automatically on alternating strokes using an air distribution valve. Air supply shall be clean dry air filtered to particle size less than 20 microns.
 - 4) Leak detection sensors shall be provided for the air chamber, which in turn trips the pump upon detection of any leak.
 - 5) It shall be possible to throttle the discharge from full flow to zero flow without pump damage. The pump shall stop automatically when the discharge is closed. Pump shall be self-priming.
 - 6) Variable flow when specified shall be achieved by regulating the air supply and the flow rate control shall be 0 to 100%.

- (j) Reciprocating Diaphragm Pumps:
- (i) For reciprocating diaphragm pumps specified for dosing pump application, it shall meet the following additional requirements:
- 1) Reciprocating diaphragm type driven by electric motors. The pump, motor and drive arrangement shall be mounted on a robust combination baseplate. Unless otherwise specified, only one liquid end shall be motor-driven.
 - 2) Pump heads shall be manufactured from thermoplastic materials suitable for the maximum concentration of chemicals to be handled.
 - 3) Variable stroke mechanisms shall be incorporated in the drive arrangement to allow infinitely variable adjustment of pump output by means of a micrometer, hand wheel or similar mechanical device whilst the pump is running.
 - 4) Where the pump is part of an automatic chlorine residual control loop, the stroke mechanism shall be fitted additionally with a three phase bi-directional motor with torque limiter and automatic stops at both extremes of travel. A position feedback potentiometer shall be provided to facilitate control and remote indication of position. The operational range of stroke adjustment shall be not less than 10:1.
 - 5) The pump head shall be driven through a totally enclosed speed-reduction gearbox with integral reciprocating drive device of the adjustable crank or mechanical lost motion type. The gearbox and reciprocating drive shall be oil bath lubricated. The unit shall incorporate filling and drain plugs for oil and a sight glass for oil level.
 - 6) Where reciprocating diaphragm pumps are hydraulically actuated, the design shall ensure that any failure in the diaphragm displacement chamber or diaphragm does not result in leakage of transmission oil into the chemical delivery system.
 - 7) Drive motors shall be of the three-phase cage induction type whether for fixed speed or variable speed operation.
 - 8) Where variable speed operation is specified, the speed turn-down ratio shall be not less than 5:1.
- (k) Peristaltic Pumps:
- (i) For Peristaltic pumps specified for dosing pump application, it shall meet the following additional requirements as minimum:
- 1) The type of which the pumping action results from alternate compression and relaxation of a machined hose between pump housing and compressing shoes. These shall be capable of operating in either rotational direction, running dry without damage to pump or hose and pulling up to 95 % full vacuum i.e. 9.5 m w.c. Pump output shall conform to achieve a repeatability of ± 1 % accuracy.

- 2) Construction of peristaltic pumps shall be of the valveless/glandless design having no dynamic seals in contact with the pumped fluid. Bearings shall also be protected from the pumped fluid by means of bearings protection external to hose chamber. A filtered breather plug and threaded drain plug shall be available for draining the pump housing.
- 3) The pump shall be capable of being rotated through 90 degree increments for four different port-mounting configurations to facilitate alternative installation orientations.
- 4) The pump casing/housing shall contain a food-grade glycerine based lubricant to lubricate the hose and provide a medium for cooling. The lubricant level and pump operation shall be visible via a removable inspection window/cover. Removal of the window/cover will facilitate shoe compression adjustment. However, hose replacement shall be achievable without cover or pump removal.
- 5) Pump construction shall include a buffer zone between gearing and pump head to prevent gearbox contamination in the event of hose lubrication seal failure. Visual detection of this failure is possible by means of the design of having the bearing cylinder venting through the rear of the pump.
- 6) The pump rotor is to be independently supported on dedicated sealed and greased-for-life ball bearings located directly under the rotor's load. The bearings are in turn supported by the bearing cylinder located within the pump housing and sealed via a static Neoprene seal. The rotor shall have two pressing shoes mounted 180 degrees apart which are adjustable for varying degrees of compression via flat shims. Rotors incorporating rollers or fixed occlusion shoes are unacceptable.
- 7) Except for small Dia. 10, 20 and 25 mm pumps, pumps of all other sizes shall have direct-coupled gear drive arrangement to combine the reliability of long-coupled pumps with the compact footprint of close-coupled pumps.
- 8) Each pump shall be supplied with steel fabricated pedestal or mounting base plate capable of carrying the weight of the pump and gear/motor and be anchored directly to the concrete foundation.
- 9) Large size pumps shall be fabricated to have BS EN 1092 flange whilst smaller pumps shall be with the proper tubing fittings or inserts.
- 10) Materials of construction for various components of peristaltic pumps shall conform to as follows:

Component	Material
Pump casing	Ductile iron ASTM A56
Mounting frame	Epoxy coated steel
Seals	Neoprene
O-rings	Neoprene
Shoes	Aluminium

Component	Material
Hose	EPDM (or alternative material guaranteed for corrosion resistance and compatibility with pumped fluid and design conditions) reinforced with multiple layers of Nylon/Kevlar
Hose inserts/fittings /connectors	Polypropylene
Flange brackets	Stainless steel
Clamps securing inserts to hose	Stainless steel SS316
Miscellaneous assembly hardware	Stainless steel

- 11) Burst hose sensor. A switch shall be located at near the top of the pump to detect leakage of pumped fluid into the housing. The corresponding sensor is to be mounted on the rear of the housing and out of the way of being knocked off. The sensor shall be supplied as Normally Closed with the provision to be adjusted to Normally Open at the field.
- 12) Tubing inserts/connectors as required shall be provided for small size pumps.

G4.4.18 Progressive Cavity Pumps

- (a) Pump Design:
 - (i) The pump shall be of the type whereby a pumping action is created by a helical rotor rotating eccentrically within a resilient stator having internally the shape of a double helical chamber. The eccentric motion of the rotor shall maintain constant seal across the stator as it travels continuously through the pump to give a uniform positive displacement.
 - (ii) The pump design shall enable the rotor, stator, coupling rod and seal to be removed and replaced in-place without disturbing or removing the suction and discharge pipe works connected to the pump.
 - (iii) Pumps shall be suitable for forward or reverse operation. That is, depending on the pipework configuration, it shall be possible to run the pump with the suction in either direction through the rotor/stator as appropriate. The pump and motor shall be mounted in line on a common baseplate assembly.
 - (iv) The pump generally shall be arranged with a single shaft seal. The shaft seal shall be of the mechanical type. On all pumps, inspection chambers shall be fitted to the casing for periodic removal of accumulated debris.
 - (v) The generation of the eccentric pumping motion shall be by either flexible drive shaft or by a rigid drive shaft/coupling rod/universal joint assembly.

- (vi) Where a rigid drive shaft is used the torque of the drive shall be transmitted through the pump drive shaft, coupling rod and rotor by two universal joints. Each universal joint shall be designed to transmit maximum torque at maximum speed and pressure and shall consist of replaceable pin and bushes manufactured from wear resistant steels. The universal joint shall be protected from the product by a totally enclosed universal joint sleeve which incorporates labyrinth type sealing at each end. Additionally, the universal joint sleeve shall be secured in place by two holding bands. The universal joint shall be lubricated with a suitable grease for its entire life.
- (vii) The pump universal joints shall be covered by a 10,000 operating hour guarantee against failure.
- (viii) Where a flexible drive shaft is provided, a corrosion resistant shroud shall be fitted around the drive shaft.
- (ix) The pump shall be driven by a fixed or variable speed electric motor as specified.
- (b) Pump Construction:
 - (i) Pump casing parts shall be manufactured from thick walled grey cast iron grade 17 to BS EN 1561. Shaft seal leakage shall be collected in the lantern drip tray, which shall be threaded to allow for a pipe connection and drainage to an external collection location.
 - (ii) The casing shall be designed so that it can be rotated through 90 degrees quadrants giving possible alternatives to positions of pump installation. The casing should also be fitted with two cleanout/inspection ports, the internal surface of which shall follow the curvature of the suction casing to prevent settlement of solids causing possible blockages. The suction and pressure casing shall be fitted with both drain and pressure/vacuum gauge connection.
 - (iii) The pump shaft shall be made of stainless steel of grade 316 at least and shall be of a single piece construction through the bearings and shaft seal area.
 - (iv) The drive shaft and rotor shall be connected by means of a rugged pin type universal joint.
 - (v) Coupling guards shall be provided on drive couplings. Guards shall be rigid, securely fixed and designed so that removal is not necessary during normal operation, routine maintenance and routine inspection.
 - (vi) The shaft bearings shall be positively isolated from fluid being pumped.
 - (vii) The helical rotor shall be manufactured from stainless steel type 316 S11 to BS EN 10216-5 and BS EN 10217-7. It shall be covered with a layer of high density hard chrome plating of at least 0.25 mm thick for abrasion resistance and surface polished to a minimum of Ra 1.9 micron to maximise the stator wear life. This shall be the minimum grade of material specification.
 - (viii) The stator shall be of the double helical moulded-to-metal type using natural or synthetic rubber or an approved alternative material. The rubber shall be bonded to a steel tube and have 10 durometer hardness. The stator shall be fastened to the suction housing and discharge flange with removable clamp rings to facilitate stator removal. The stator shall be designed to prevent the material being pumped from contacting the stator bonding and tube.

- (ix) Unless otherwise specified and approved by the S.O., the maximum rotor speed shall not exceed 400 rpm while the normal rotor speed range for progressive cavity pumps handling fluids with more than 1% solids shall be in the region of 150 RPM and the maximum speed shall not exceed 200 rpm. Suitable geared motor or gear drive shall be used to achieve this.
- (x) The pump shall be provided with a single mechanical seal. Seal shall not require external flushing. All seal hardware shall be Type 316 stainless steel. The seal faces shall be silicon carbide.
- (c) Bearings:
 - (i) At least two anti-friction bearings shall be provided. The inboard bearing shall carry radial loads, and the outboard bearing shall carry both radial and thrust loads. Each bearing shall have an AFBMA L10 Life Rating of 100,000 hours at the maximum differential pressure. Shaft speed shall not exceed the limits specified by the bearing manufacturer.
- (d) Gears:
 - (i) The gears shall be designed and manufactured in accordance with AGMA Standards, with a minimum service factor of 2 for the Cycloidal gear and 1.5 for the Helical, spiral bevel, combination bevel-helical, and worm gear based on the nameplate power rating of the drive motor and a minimum efficiency of 94 percent.
 - (ii) Gear speed reducers shall be of the enclosed type, oil or grease lubricated and fully sealed, with a breather to allow air to escape but keep dust and dirt out. The casing shall be of cast iron or heavy-duty steel construction with lifting lugs and an inspection cover for each gear train. An oil level sight glass, oil flow indicator, and oil drain shall be provided and installed for easy access.
 - (iii) Material selections for gear shall comply with AGMA values and the manufacturer's recommendations. Input and output shafts shall be properly designed for the service and load requirements. Gears shall be computer-matched for minimum tolerance variation. The output shall have two positive seals to prevent oil leakage.
- (e) Pump Protection:
 - (i) The pumps shall be fitted with individual dry running protection device to initiate pump trip for prevention of possible damage to the stator/rotor. Dry running protection by 'under current' monitoring or 'pipeline intrusive' devices shall not be used.
 - (ii) The pump shall be fitted with either an electronic or mechanical means of over pressure protection system. The electronic system shall consist of an over-pressure sensing device able to detect high pressure in the delivery pipework immediately downstream of the pump. The device shall be able to initiate pump trip over an adjustable range of 50 to 300% above normal operating pressure. Means shall be provided to prevent trip due to short time pressure transients. The mechanical system shall consist of a pressure relief valve fitted to the discharge line of the pump with fluid that is relieved piped back to the suction side of the pump.

- (iii) Either electronic or mechanical means of over pressure protection system shall not come into contact with the fluid being pumped.
- (iv) A temperature sensor (thermistor sensor) shall be provided in the stator of the pump to measure the temperature between stator and rotor. If the stator temperature rises above the intended cut-off temperature because of heating due to dry running, a signal indication shall be shown in the control panel and the pump shall be stopped immediately. The set-point of the cut-off temperature shall be adjustable. The preferred method to monitor the temperature between the rotor and stator is by a thermistor sensor fitted to the stator.

G4.4.19 Screw Induced Flow Centrifugal Pump

- (a) These pumps are meant for pumping either sludge, aerated effluent or plant drain application which could contain long fibrous material and solid particles.
- (b) Pump Design:
 - (i) Each pumping unit shall be complete with a pump, electric motor, coupling, coupling guard, anchor bolts, and other appurtenances specified or otherwise required for proper operation, all mounted on a common baseplate. Pumps of the close-coupled type with the impeller attached directly to the motor shaft without pump bearings or flexible couplings will not be acceptable.
 - (ii) The basic design shall be a single-passage, vertical spindle, clog-free pump utilising a screw-centrifugal impeller. The overall pump design shall combine high efficiency, low required NPSH, a large solid passage, the ability to handle rags or other fibrous material without plugging and the ability to handle high solids concentrations efficiently.
 - (iii) Design shall permit low liquid velocities, gradual acceleration and change of flow direction of a pumped media. Impeller/casing design shall result in a single helical passage free of surfaces to which solid or fibrous material can adhere.
 - (iv) Unless otherwise specified, the impellers shall be made of stainless steel grade 316L as a minimum or better and be smooth, well finished, free from blowholes and imperfections. The size of the impeller shall not be larger than 98% of the full impeller size for the particular size of the pump.
 - (v) A single flow stream through the impeller shall be provided to ensure that raw water and any other foreign particles of stringy, paper and plastic materials are passed through the impeller without clogging or getting caught.
 - (vi) Impellers shall be both statically and dynamically balanced.
- (c) Pump Construction:
 - (i) The pump casing shall be of grey cast iron ASTM A48 Class 30 or better with ductile iron as an acceptable alternative. It shall be of sufficient strength, weight and metal thickness to ensure long life, accurate alignment and reliable operation. A minimum 100 mm diameter clean-out opening with a removable cover shall be located on the casing to allow for inspection and access to the impeller.

- (ii) The pump shall be provided with a suction liner and the same shall be with a spiral groove on the inside to avoid clogging of fibrous material between impeller and pump suction cover. The material of construction for the suction liner shall be abrasion resistant Ni-hard or 27 percent chrome cast iron conforming to ASTM A532 with a minimum Brinell hardness of 450.
- (iii) All bolts, nuts and washers shall be type SS 316.
- (iv) The volute casing and bearing frame housing shall be fitted with all necessary drain, discharge pressure gauge, flushing, and lubrication service connections.
- (v) The bearing housing shall be of cast iron, ASTM A48CL30 or ductile iron and shall be of the back pull-out design so that the bearing housing and impeller can be removed without disconnecting the casing from the suction and discharge piping. Bearings shall be of the ball or roller type. The heavy-duty bearing frame shall be specifically designed for a L-10 bearing life of 50,000 hours. Bearings shall be grease lubricated with provision for addition and relief of the grease
- (vi) Pump shaft shall be accurately machined and be of sufficient strength to transmit full driver output. The material of the shaft shall be Stainless Steel grade 420 or better.
- (vii) The pump shall be provided with double mechanical seal (tandem combination) in a special bearing housing frame adapter installed on the pump shaft. The mechanical seals shall be fitted with one half running in the liquid being pumped and the other in an oil bath between the back cover of the pump and the bearing frame. The seal faces shall be Carbon/Ceramic. All metal parts of seal shall be Type 316 stainless steel.
- (viii) The pump suction bend, on vertical pumps, shall be provided with an inspection hatch and cover.
- (ix) The vertical dry pit pump shall be supported on a pedestal base with openings large enough to permit access to the suction line and to the inspection hatch in the suction line. The base shall be rugged enough to support the full weight of the pump and motor. The motor weight shall not be transferred to the pump directly. The motor load shall be transferred to the pedestal base supported by a separate structure.
- (x) Horizontal pumps shall be provided with a flexible spacer type coupling and safety guards. The horizontal pump and motor shall be mounted on a common fabricated steel base.
- (xi) A seal flushing provision using effluent water or better grade water shall be provided for seal flushing before start-up of pump. The design pressure shall be 0.5 bar above the operating pressure.
- (xii) Glycerine filled 100 mm discharge gauge with diaphragm suitable for sludge shall be provided.

G4.4.20 Chopper Pumps

(a) Design Requirement:

- (i) The pump shall incorporate a rotary cutting mechanism. This shall be suitable to cut raw sewage (used water) or sludge solids and screenings into pieces small enough to pass through the impeller.
- (ii) The pump components which incorporate the cutting edges shall be of hard tool steel, minimum Ni Resistant ASTM A439.80 Type D2 combined with hardened white alloy iron cutter plate or equivalent or better. Provision shall be made in the pump construction for the easy replacement of all cutting components.
- (iii) Casing shall be volute design and manufactured from ductile cast iron with all water passages to be smooth, and free of blowholes and imperfections for good flow characteristics. For end suction type it shall be back pull design with jacking bolts for adjustment of impeller to cutter bar clearance and shall allow removal of pump components. For submersible shall be closed coupled and suitable for dry well installation application.
- (iv) The casing shall have provision of drain pipe with isolation valve to facilitate the draining of the pumped fluid prior to casing dismantling.
- (v) All exposed nuts or bolts shall be stainless steel 316 as minimum.
- (vi) Impeller shall be multi vane semi-open type with replaceable cutting head.
- (vii) Impeller shall be cast alloy steel heat treated and dynamically balanced.
- (viii) The impeller through let shall be suitable to pass macerated solids of the size produced by the cutting action of the pump.
- (ix) The impeller shall be secured to the shaft using a cutter nut or threaded directly to pump shaft and be designed to cut stringy materials and prevent binding using a raised, rotating cutter tooth. The cutter nut shall be cast steel. If the impeller is threaded directly to the pump shaft, it shall be such that the cutter bar can extend across the full diameter of the pump suction opening allowing the impeller to shear against the cutter bar throughout the full diameter of the suction opening, including at the center.
- (x) The impeller shall be keyed to the shaft and shall have no axial adjustments and no set screws.
- (xi) The design for the impeller and cutting plate shall prevent blockage at the intake opening and wrapping of debris at the shaft area.
- (xii) Chopper pumps utilizing individually mounted shear bars shall not be acceptable. Cutter bar shall be alloy steel as minimum.
- (xiii) The upper cutter teeth are positioned as closely as possible to the centre of shaft rotation to minimize cutting torque and motor tripping. Upper cutter shall be cast alloy steel.
- (xiv) The pump shaft and impeller shall be supported by ball bearings. Shafting design shall have no deflection during solids chopping.

- (xv) Bearing Housing shall be ductile cast iron, and machined with piloted bearing fits for concentricity of all components. Piloted motor mount shall firmly align motor on top of bearing housing.
- (xvi) The mechanical seal shall be a cartridge-type mechanical seal. This cartridge seal shall be pre-assembled and pre-tested so that no seal settings or adjustments are required during installation on site. Any springs used to push the seal faces together must be shielded from the fluid to be pumped.
- (xvii) An oil level switch shall be mounted at the top of the wet well, with a hose feeding down to the side of the bearing housing to detect oil level and shut off the motor in event of low oil level. A sensitive relay shall be included for mounting in the motor control panel.
- (xviii) The submersible motor if required, shall be close coupled directly to the pump shaft using a solid sleeve coupling, which is keyed to both the pump and motor shafts. Slip clutches and shear pins between the shaft and the motor are not acceptable.

G4.4.21 Rotary Lobe Pumps

- (a) General:
 - (i) The pumps shall have an output characteristic which is independent of change in heads.
 - (ii) The pumps shall operate on the Roots' principle, using lobed rotors mounted on two counter-rotating shafts. The design shall be optimized for handling sewage sludge containing suspended solids, screenings, and grit.
 - (iii) The rotors shall be driven by gears and shall operate without rubbing contact.
 - (iv) Gearbox used shall be ruggedly designed to withstand the operating conditions.
 - (v) The pumps shall be self-priming and shall have a 25 mm tapping on the suction side for manually flushing the pumps with plant/service water. The pumps shall have safety features incorporated to prevent dry running.
 - (vi) The pumps shall be designed to handle sludge and the Net Positive Suction Head (NPSH) required for the pumps when pumping singly or in any combination as specified shall satisfy the NPSH available margin. The corresponding calculations are to be submitted by Contractor for verification.
- (b) Pump Construction:
 - (i) The pump casing shall be ruggedly constructed from ductile iron. The rotors shall be tri-lobe type and shall be moulded with urethane or natural rubber. Notwithstanding the above, the rotor material shall be compatible with the fluids being pumped to ensure trouble free operations.
 - (ii) The rotors shall be in a helical or other configuration designed to reduce pulsations to a minimum.
 - (iii) Pump rotors shall have a moulded, elastomer covering bonded onto a metal insert.
 - (iv) The rotor design shall incorporate elastomer lobes or lobe tips to reduce wear from grit and other abrasives.

- (v) The rotor shape shall allow free passage of sewage solids including rags.
 - (vi) The pump casing shall have an easily removable door or hatch allowing access to the rotors. The rotors shall be designed to be easily withdrawn and replaced through this opening, with the use of special tools if required.
 - (vii) Suction and discharge branches shall be arranged in an inline horizontal plane.
 - (viii) All bearings shall be radial or thrust bearing and shall be lubricated by oil.
 - (ix) When the size of suction and discharge nozzle are 80 mm or larger, the pump rotor shall be provided with a rotor torque locking retention device.
 - (x) Bearing temperature, stator temperature and vibration monitoring instrumentations shall be provided.
- (c) Gearbox:
- (i) The rotor shafts shall be geared together with a robust gear train. The gears shall be designed to withstand shock loadings and jamming due to the difficult nature of the process fluid. There shall be sufficient design margin that the gear tolerance shall preclude rubbing contact between the rotors, even after long service. The gearbox shall be designed for loadings calculated according to the relevant parts of ISO 6336: 2006, and the expected service life (Part 6) shall be at least 20 years. It shall have an AGMA rating of 2.
- (d) Sealing:
- (i) The pumps shall be equipped with double seals separated by an oil bath. The main seal shall have both faces of either silicon carbide or tungsten carbide. There shall be provision for the installation of a leakage monitor if required.
- (e) Pumped Fluid:
- (i) This shall be sewage sludge as defined in the general clauses for sewage pumps. Dry solids concentration shall be up to 15 %.
- (f) Pressure Relief Valve:
- (i) Each pump shall be supplied with an appropriately rated pressure relief pinch valve together with the pipework and fittings.

G4.4.22 Recessed Impeller Pump

- (a) The pumps shall be of the fully recessed, slurry type design, with the impeller mounted completely out of the flow path between the pump inlet and discharge connections, so that solids are not required to flow through the impeller.
- (b) Pump Design:
 - (i) The Pumps shall be specifically designed to pump slurries of grit, debris and organic solids.
 - (ii) The pump casing shall be of the two-piece, radially split-type, with a separate and removable suction piece designed so that the impeller can be withdrawn without the need to remove the discharge casing or disturb the discharge piping.

- (iii) The hydraulic design of the impeller shall be such that fluid flow is directed to the suction piece, which shall be designed as a sacrificial part that is easily replaceable. The impeller shall be of the cup type design having the vane ends surrounded by an integral shroud. Impellers of the radial design that incorporate the impeller in a recessed portion of the volute or wear plate are not acceptable.
 - (iv) A removable wear plate shall be provided at the back of the impeller designed to direct flow from behind the impeller to the centre of the volute for maximum protection to the casing.
 - (v) A single cartridge mechanical seal which requires no external flushing shall be provided for the pump. The seal chamber shall be specifically designed to minimise the contact between the grit and the seal faces. The mechanical seal shall have a minimum guaranteed life of 2 years.
- (c) Mechanical Seal:
- (i) Contractor shall suitably design the mechanical seal suitable for handling abrasive medium (grit, etc.) and sewage application.
 - (ii) The mechanical seal shall be of the type that is self-adjusting and without external flushing. It shall have independent self-aligning, rotating and stationary parts. The wear surfaces of the seal must be removable from the seal for inspection or be repaired easily without any special tools.
 - (iii) The rotational sealing ring shall be tungsten carbide and stationary sealing ring shall be silicon carbide material, with flatness achieved by lapping process. The spring material shall be heat treated carbon steel ASTM A-682 or equivalent and shall be totally encapsulated in a stainless steel (Type SS 316) or equivalent to protect from the pumped medium.
 - (iv) The mechanical seal assembly shall be mounted in a Hi-Chrome iron, ASTM A-532, steel seal chamber in such a way that pumped medium will contact and lubricate the seal faces. The seal shall be installed into the seal chamber from the impeller side of the pump so that only the casing/suction piece and impeller need to be removed to access the seal for seal maintenance/replacement.
 - (v) The seal chamber shall be specifically designed to minimise the contact between the grit and the seal faces. The mechanical seal shall have a minimum guaranteed life of 2 years.
- (d) Pump Construction:
- (i) As a minimum, the material of construction for the pump shall be as follows:

Pump Casing	Ni-Hard ASTM A-532, 650 Brinell hardness
Impeller	Ni-Hard ASTM A-532, 650 Brinell hardness
Wear Plate	Ni-Hard ASTM A-532, 650 Brinell hardness
Shaft	High Tensile Steel AISI 1045
Shaft Sleeve (if required)	Ni-Hard ASTM A-532, Type 416
Pump Frame/ baseplate	Structural Steel

- (ii) The wear-liners shall be of the easily replaceable type and the clearance between the impeller and the casing shall be easily adjustable without dismantling the pump.
 - (iii) Where necessary and applicable, the pump shaft shall be protected by a removable hardened stainless steel sleeve of material as specified above. Pump shaft shall be accurately machined and be of sufficient strength to transmit full driver output.
 - (iv) The impeller shall be balanced both statically and dynamically to have smooth running and shall be removed easily from the shaft for maintenance purposes.
 - (v) Belts and sheaves to drive the pump at the speed necessary to meet the specified conditions shall be provided and installed. Safety guard of galvanised steel or reinforced fibreglass shall be provided for any rotating parts.
- (e) Bearings:
- (i) Bearings shall be oil bath lubricated. The oil reservoir shall be sealed at both ends to prevent entrance of foreign matter. The thrust bearings shall consist of three angular contact ball bearings for maximum protection from all thrust loads. The bearing housing shall be equipped with a pressure venting device oil fill tap, and drain tap. A built-in sight glass shall be supplied to check proper oil level. The bearings shall be rated for a minimum L-10 life of 50,000 hours, without credit for any rear pump out vanes to balance hydraulic thrust.
- (f) Pump Assembly:
- Unless otherwise specified, the following are minimum requirements for the pump assembly:
- (i) For horizontal pumps, the common pump and motor base shall be manufactured from minimum 10 mm thick fabricated steel, suitably constructed to support the full weight of pump and motor. Horizontal pumps shall be provided with a suitable drive arrangement such as mechanical coupling or V-belt drive between motor and pump. The motor shall be mounted on an adjustable motor base so that the motor can be easily moved for V-belt tensioning and adjustment.
 - (ii) For vertical dry well pump, the motor shall be connected to the pump by means of a flexible spacer coupling with fabricated steel or reinforced fibreglass safety guard. The pump shall be supported on a pedestal base with openings large enough to permit access to the suction line and to the inspection opening in the suction elbow. The base shall be rugged enough to support the full weight of the pump and motor, and the legs of the pedestal shall be of such a length that the suction elbow of the pump will not touch the floor or level foundation upon which it stands.

- (iii) For wet well installation, the pump shall be close coupled with the motor. The pump assembly shall include a slide away coupling system. The slide away coupling assembly shall consist of a foot-mounted discharge elbow and adapter, steel baseplate, upper and lower rail supports, lifting yoke and cable. All metal to metal interfaces where movement might occur shall be non-sparking. The foot mounted discharge elbow and adapter shall conform to ASTM A48 Class 25 or 30 grey iron. The baseplate shall be provided with adjustable stops to support the pump weight in order to eliminate unnecessary strain on the coupling. Coupling shall be provided with gasket and be adjustable to eliminate leakage.

G4.4.23 Pump Manufacturer's Services

The Contractor shall engage the pump manufacturer to conduct inspection, testing, commissioning, and field adjustment as follows:

- (a) Manufacturer's service personnel shall be present in the field full-time during installation and commissioning to inspect, verify, commission, conduct field adjustments and to certify in writing that the pumps, power termination, controls and all associated appurtenances have been properly installed, aligned, lubricated, adjusted, and readied for operation with all necessary safety features incorporated.
- (b) The Contractor shall have provision for a minimum of 5 days of checking and commissioning service by the vibration and condition monitoring equipment manufacturer.
- (c) The Contractor shall have provision for a service representative of the mechanical seal manufacturer to inspect and certify the correct type of seal, correctness of installation and proper functioning of the seals.

G4.4.24 Pump Tests (Off Site and Factory Test)

- (a) Each Pump shall be tested at the factory for capacity, power requirements, efficiency at the specified duty points, Net Positive Suction Head (NPSH), discharge head, shutoff head and operating head extremes at as many other points as necessary for accurate performance curve plotting.
- (b) All tests and test reports shall be made in conformity with the requirements and recommendations of the BS EN ISO 9906 or equivalent approved by the S.O..
- (c) Acceptance testing shall be with no minus tolerance or margin allowed, unless otherwise specified by the S.O..
- (d) The pumping unit shall be shop tested with the motor to be installed in the work. For pumping units with variable frequency drives, the wire-to-water efficiency test shall include the VFD.
- (e) Detailed string test requirements shall be according to the requirements specified in the General Specification for electrical works.
- (f) The S.O. and/or the S.O. representative may witness the shop tests, inspect and check the testing equipment used, and observe the calibration of pressure gauges and transducers.

- (g) The Contractor shall furnish to the S.O. a plan and elevation drawing of the test setup showing the piping and instrumentation for approval.
- (h) The Contractor shall notify the S.O. at least 10 days in advance of the time of each shop test.
- (i) For witnessed tests, all readings are to be read manually from the certified and/or calibrated instruments. The use of computer data acquisition systems will not be acceptable.
- (j) The pump shall be verified against all specified operating points and/or operation envelope without hydraulic deficiencies.
- (k) If the pump fails to operate properly or fails to meet the specified conditions or requirements during witnessed shop testing, the Contractor shall ensure that the pump manufacturer modify the pumping unit and perform additional tests.
- (l) All labour, travel and expenses incurred by the Board for the witnessing or observation of additional tests, whether due to incomplete tests or re-testing, shall be borne by the Contractor.
- (m) All certified test reports shall be prepared and submitted to the S.O.. The Contractor shall ensure that five certified copies of a report covering each test shall be prepared by the pump manufacturer and delivered to S.O. not less than 10 days prior to the shipment of the equipment from the factory.
- (n) The report shall include data and test information as stipulated in the BS EN ISO 9906 standard. Copies of the original test logs, test reading to curve conversion equations, and certified performance curves shall be submitted to the S.O..
- (o) The curves shall include head, pump input power, pump efficiency, wire-to-water efficiency (when specified), RPM and shop test NPSH available plotted against capacity. All curves shall be easily readable having all test points clearly shown and plotted to scales consistent with performance requirements.

G4.4.25 Preparation Prior Installation

- (a) The Contractor shall engage the pump manufacturer to inspect and verify at site that the locations, spaces, elevation levels, sleeves or box-out holes, piping sizes, elevations and orientations and any entity that affect pump setting are correct during their construction by the Contractor. Additionally, the Manufacturer shall inspect the Site at least 30 days before installation of the equipment to verify that the Site is ready for pump installation works.
- (b) After hydrostatic or other tests, all entrapped fluid shall be drained prior to shipment and proper care shall be taken to protect parts from the entrance of water during shipment, storage, and handling.
- (c) No shipment shall be made until approved by the S.O. in writing.
- (d) One year's supply of the manufacturer's recommended lubricating oil, grease required for the equipment operation shall be delivered with the pumps and motors.

G4.4.26 Field Installation Quality Control, Testing and Commissioning

The minimum requirements for the site installation and commissioning shall be as follows:

- (a) Installation Supervision on Site
 - (i) The Contractor shall submit the details of his equipment manufacturer's qualified field installation supervisor particulars during the equipment installation.
 - (ii) The Contractor shall engage the Manufacturers' installation supervisor to observe, instruct, guide and direct the Contractor's erection or installation activities for pumps.
 - (iii) The Contractor shall give written notification to the Manufacturer 10 days prior to the need for the services of the Manufacturer's installation supervisor.
- (b) Installation Check
 - (i) The Contractor shall engage an experienced, competent and authorised representative of the manufacturer to visit the site of the work to inspect, check, adjust if necessary and approve the equipment installation.
 - (ii) The Contractor shall assess and implement the means of protecting and maintenance of pumps that are installed prior to full structural completion. The installation schedule for different components of the pumps eg. motor, shaft, pump casing, etc. shall be carefully planned out so as to be well coordinated with the structural works.
- (c) The representative shall be present when the equipment is placed in operation in accordance with manufacturer's start up requirements, revisit the job site as often as necessary until all troubles are corrected and the equipment installation and operation are satisfactory in the opinion of S.O..
- (d) The Contractor shall furnish the manufacturer's written report certifying that the equipment has been properly installed and lubricated; is in accurate alignment; is free from any undue stress imposed by connecting piping or anchor bolts; and has been operated under full load conditions satisfactorily.
- (e) Testing of Pumps at Site
 - (i) All site tests shall be witnessed by the S.O.. The Contractor shall notify the S.O. in writing two (2) weeks in advance of each site test to request for the witnessing.
 - (ii) After installation, the pumps, motors, transmitting shafts bearing housing brackets, etc. shall be:
 - 1) Rigidly bolted down to the supporting plinth, foundation, etc.
 - 2) Properly and accurately aligned; and
 - 3) Operated freely without noticeable noise and vibration.
 - (iii) Rotodynamic pumps shall be tested in accordance to BS EN ISO 9906 standard and this specification.

- (iv) Positive displacement pumps shall be tested in accordance to BS EB 14343 and this specification.
 - (v) Measurements shall be recorded for the supply voltage, amperage, pump suction head and pump discharge head for at least 4 pumping conditions at each pump rotational speed.
 - (vi) Each power lead to the motor for proper current balance shall be checked.
 - (vii) All pumps' performance, noise, vibration, design and condition shall be tested and verified.
 - (viii) The pump shall be verified against all specified operating points and/or operation envelope without hydraulic deficiencies.
 - (ix) The maximum bearing temperature for pumps and motors shall not be more than 40 deg. C. above the ambient temperature.
 - (x) Under normal operating conditions, the pumping equipment shall be capable of:
 - 1) Pumping continuously and reliably without requiring continuous surveillance; and
 - 2) Operating quietly, without noticeable vibration and without cavitation, single or in any combination from minimum to maximum operating fluid levels specified or operating against closed delivery valve at shut off head.
 - (xi) The S.O. shall reject any pump set during the tests if it:
 - 1) Fails to pump under normal operating conditions reliably;
 - 2) Exceeds the vibration criteria limits as specified in BS ISO 20816-1 or as specified in the Particular Specification;
 - 3) Shows signs of cavitation; or
 - 4) Fails to comply in any other respect with the requirements of this specification; and
 - 5) Exhibits any defect or damage which in the S.O.'s opinion warrants its rejection.
 - (xii) Expenses for the arrangement of water to carry out the tests shall be allowed for in the contract sum/tender prices.
- (f) Commissioning Test on Pump Sets:
- (i) Trial Run:
 - 1) Rotation direction check and 30 minutes of free load run shall be conducted for each main motor before coupling to the main pump. Each main pump set shall then run singly at the principal duty point for about 1 hour during which the operation of non-return valve and emergency stop will be checked.

- (ii) Load Run:
 - 1) The new pump sets shall be run individually and in multiples, based on specified control philosophy, for a continuous duration of at least eight (8) hours unless otherwise specified and agreed with the S.O.. The proposed sequence shall be submitted for the approval of the S.O.. If the pump set trips due to whatever fault during the agreed duration, the test shall start afresh after the defect has been rectified. If these tests show any noticeable deviation in the performance compared with the works test results, etc, the Contractor shall immediately take steps to rectify the defects. The Contractor shall ensure that at least one competent engineer is present throughout the duration of the above tests.
 - 2) The Contractor shall submit the Test Schedule, Test Procedures, test results recording formats and six (6) copies of the Operation and Maintenance Instruction Manuals to the S.O. for approval at least 2 weeks before the tests. Any test shall not be commenced prior this submission completion.
- (iii) The Contractor shall supply all personnel, tools, instruments, materials and other equipment necessary for carrying out the tests. All measuring instruments shall be suitably graduated, calibrated and certified by an approved independent testing authority prior to the tests.
- (iv) The Contractor shall supply and wire up one set of precision ammeter, voltmeter and wattmeter with an accuracy of better than + 1%.
- (g) Performance Test for Pump set by using a power monitoring method:
 - (i) The performance test for pumps by the power monitoring method, if required, shall conform to the following requirements as a minimum:
 - 1) The Contractor shall carry out the tests during commissioning to verify the factory test curves of the concerned pumps. The maximum deviation allowable from the factory test curves for flowrate and efficiency at the duty head(s) shall be $\pm 5\%$;
 - 2) The data collected during the tests shall be used to plot the total head efficiency and motor input power versus flow rate curves for the pumps;
 - 3) 4 copies of the completed test report shall be submitted within 1 week of the performance tests;
 - 4) The Contractor shall supply all personnel, tools, instruments, materials and any other equipment necessary for carrying out the tests;
 - 5) The Contractor shall fabricate and install all necessary fittings and isolation valves at the suction and delivery pipework of the pump sets for the sensing devices required for the test; and
 - 6) The cost for carrying out the tests shall be included in the Tender Price.
- (h) For each successfully completed site test, the Contractor furnish the test records and certifications stating the test date, test data and test result with a listing of all persons present during the test. The certification shall also record all final adjustments that have been made to the tested pump(s).

END OF SECTION